

# Gravitational Lensing as Phase Refraction

*A Derivation of All Standard Lensing Observables from the Phase-Cost Variational Principle, Without Invoking Spacetime Curvature as a Primitive*

Marlon Hanks

*Independent Researcher, Dust LLC*

*Correspondence: [email protected]*

arXiv:2601.XXXXX [gr-qc, hep-th] Submitted 16 January 2026

Received: 16 January 2026 • Revised: — • Accepted: —

**Keywords:** gravitational lensing, phase refraction, Phase Theory, admissibility index, phase-cost variational principle, emergent metric, weak lensing, strong lensing, gravitational waves, photon sphere, Shapiro delay, dark matter, Einstein ring, caustics

**MSC2020:** 83C30, 83C57, 78A05, 53Z05 **PACS:** 98.62.Sb, 95.30.Sf, 04.20.Cv

## Abstract

We present a comprehensive derivation of gravitational lensing entirely within the framework of Phase Theory (Hanks 2026), demonstrating that all standard lensing observables—light deflection angles, Shapiro time delay, convergence maps, shear fields, Einstein ring radii, magnification, multiple imaging, caustic structure, and time delay cosmography—emerge naturally from the phase-cost variational principle without invoking spacetime curvature as a primitive quantity. In Phase Theory, massive objects are stable topological defects in a

pre-geometric phase substrate  $\Phi: M \rightarrow T$ , and their gravitational influence is encoded in spatial gradients of the admissibility index  $\eta(x) = \sqrt{(K_0/K(x))}$ , which plays the role of a spatially varying refractive index for massless phase modes (light). The equation governing phase-mode propagation,  $\delta f_\gamma \eta(x) ds = 0$ , is derived from the Euler-Lagrange equations of the linearized phase-inconsistency functional in the massless sector. At first post-Newtonian order, this reproduces every prediction of general relativistic lensing theory identically, including the canonical factor-of-two enhancement of the Newtonian deflection. The temporal component of the admissibility gradient, rooted in Axiom 2 (Update Ordering), contributes exactly half the total deflection, providing a conceptually transparent explanation for a result that has historically required the full apparatus of Riemannian geometry. Beyond first order, Phase Theory predicts coherence-suppressed magnification near caustics, a high-multipole cutoff in the weak lensing power spectrum from neutral topological sectors (dark matter analogs), a waveform asymmetry in strongly lensed gravitational waves, and a measurable correction to the photon ring angular diameter accessible to next-generation Event Horizon Telescope observations. We provide full derivations, comparison tables, experimental protocols for seven discriminating tests, and a complete reference list. This paper is Paper 45 of the Phase Theory series.

## Contents

1. Introduction and Scope
2. Phase Theory Foundations Relevant to Lensing
3. Massless Phase Modes and Light
4. Phase Density: Formal Definition and Properties
5. The Admissibility Index

- 6.** Admissibility Gradients and Their Origin
- 7.** Phase Refraction: The Fundamental Mechanism
- 8.** The Phase-Cost Variational Principle
- 9.** Derivation of the Deflection Angle
- 10.** The Temporal Component: Ordering Cost
- 11.** Gravitational Time Delay (Shapiro Effect)
- 12.** Lensing Geometry in Phase Theory
- 13.** Convergence and Shear in Phase Geometry
- 14.** Multiple Imaging: Topology of Admissible Paths
- 15.** Caustics and Critical Curves
- 16.** Magnification Without Energy Gain
- 17.** Weak Gravitational Lensing
- 18.** Strong Gravitational Lensing
- 19.** Einstein Rings
- 20.** Cluster Lensing
- 21.** Dark Matter as Lensing Source
- 22.** Microlensing
- 23.** Quasar Microlensing and Flux Anomalies
- 24.** Lensing of the Cosmic Microwave Background
- 25.** Time Delay Cosmography
- 26.** Refractive Gravity Deviations Near Extreme Admissibility
- 27.** Phase Theory Lensing vs. Modified Gravity Theories
- 28.** The Refractive Index Tensor for Anisotropic Lenses

- 29.** Wave Optics and Coherence in Phase Lensing
- 30.** Gravitational Wave Lensing as Phase Lensing
- 31.** Plasma Lensing and Dispersive Effects
- 32.** Self-Lensing and Phase Feedback
- 33.** Lensing Selection Effects and Survey Biases
- 34.** Precision Tests of the Lensing Deflection Law
- 35.** Predictions Unique to Phase Theory Lensing
- 36.** Experimental Protocols for Phase Lensing Tests
- 37.** Comparison with General Relativity: Formal Equivalence and Deviations
- 38.** Implications for the Ontology of Gravity
- 39.** Appendix A — Phase-Cost Geodesic Equations
- 40.** Appendix B — The Admissibility Transfer Equation
- 41.** Glossary of Phase Lensing Terms
- 42.** References

---

## **1. Introduction and Scope**

Gravitational lensing stands as one of the most dramatic and epistemically consequential predictions of general relativity. When Arthur Stanley Eddington and his collaborators measured the deflection of starlight during the solar eclipse of 29 May 1919, the result—approximately 1.75 arcseconds for a ray grazing the solar limb—confirmed Einstein's theory at the expense of Newtonian gravity and elevated curved spacetime from mathematical conjecture to empirical reality. The experiment was, and in the popular imagination remains, the definitive demonstration that gravity bends space. For over a century, every precision test of gravitational lensing has been interpreted through the language of Riemannian geometry: null geodesics

trace the curved spacetime manifold, and the deviation of light from its Euclidean path is a kinematic consequence of the curvature of the metric tensor  $g_{\mu\nu}$ .

Yet the success of this geometric interpretation has never constituted a proof of its ontological necessity. That a physical phenomenon can be derived from curved spacetime does not logically entail that curved spacetime is the only—or even the most fundamental—explanation for that phenomenon. The history of physics is replete with situations in which a successful formal description (Ptolemaic epicycles, caloric fluid, the luminiferous aether) was later understood as an effective description of a deeper mechanism. The present paper examines whether gravitational lensing belongs to this category: specifically, whether lensing observables can be fully derived from a geometry-free theory of gravity, and what conceptual clarification such a derivation provides.

The framework adopted here is Phase Theory, developed in the preceding papers of this series (Hanks 2026, Paper 1). In Phase Theory, the primitive object is not a spacetime manifold equipped with a metric, but a pre-geometric topological space  $M$  carrying a phase substrate  $\Phi: M \rightarrow T$ , where  $T = U(1) \times SU(2) \times SU(3)$  encodes the full gauge structure of the Standard Model. The metric tensor is a derived quantity (Axiom 5), emerging from the two-point correlator of phase-field gradients. Gravity is not the curvature of spacetime; it is the spatial variation of the admissibility index  $\eta(x)$ , which measures the local cost of phase reconfiguration relative to the vacuum. Massive objects are stable topological defects in the phase substrate that elevate the local admissibility index in their vicinity. Massless phase modes—the Phase Theory description of light and other zero-rest-mass quanta—propagate along paths that minimize the integrated admissibility cost, just as light propagates along paths of stationary optical path length in a medium of varying refractive index.

The thesis of this paper is precise and falsifiable: every lensing observable predicted by general relativity at first post-Newtonian order is reproduced identically by Phase Theory's phase-cost variational principle, and Phase Theory further predicts a set of higher-order corrections and novel signatures that are experimentally discriminable from GR using current and near-future instrumentation. The derivation requires no curved spacetime, no Riemann tensor, no parallel transport, and no equivalence principle as a postulate. Instead, it requires only the five axioms of Phase Theory, the

definition of the admissibility index, and the calculus of variations applied to the phase-cost functional. The conceptual payoff is substantial: lensing, long regarded as the canonical exhibit of the reality of curved spacetime, turns out to require only a physical medium with a spatially varying propagation cost—a profoundly more intelligible phenomenon.

The paper is organized as follows. Sections 2–8 establish the theoretical foundations: Phase Theory axioms, the nature of massless phase modes, the definitions of phase density and admissibility index, the origin of admissibility gradients near stable defects, and the phase-cost variational principle governing propagation. Sections 9–11 carry out the core derivations: the deflection angle (full GR result, factor-of-two enhancement included), the temporal ordering contribution, and the Shapiro delay. Sections 12–19 develop the full lensing formalism in Phase Theory, including the lens equation, convergence and shear, multiple imaging, caustics, magnification, weak lensing, strong lensing, and Einstein rings. Sections 20–25 apply the formalism to specific astrophysical regimes: cluster lensing, dark matter, microlensing, quasar microlensing, CMB lensing, and time delay cosmography. Sections 26–36 address deviations from GR, comparisons with other modified gravity theories, anisotropic lenses, wave optics, gravitational wave lensing, plasma lensing, self-lensing, selection effects, precision tests, and unique predictions. Sections 37–38 provide the formal equivalence theorem and its proof, and a discussion of the ontological implications. Appendices A and B give detailed mathematical derivations. A Glossary and full Reference list complete the paper.

---

## **2. Phase Theory Foundations Relevant to Lensing**

Phase Theory rests on five axioms that collectively replace the postulates of general relativity and quantum field theory with a single substrate-based framework. We review here the axioms in a form adapted to the propagation problems central to gravitational lensing, emphasizing those structural features that differ most sharply from the geometric picture of GR. For full statements and motivations, the reader is directed to Paper 1 of the series (Hanks 2026).

**Axiom 1 (Global Phase Consistency)** states that the realized phase configuration  $\Phi^*$  is the global minimizer of the phase-inconsistency functional  $I[\Phi]$  over the space  $\Gamma$  of all admissible phase fields:  $\Phi^* = \arg \min_{\Phi \in \Gamma} I[\Phi]$ , with  $I[\Phi] \geq 0$ . The functional  $I[\Phi] = \int_M \rho(\Phi, D\Phi) \, d\mu$  encodes the total cost of phase misalignment across the substrate, where  $\rho = \frac{1}{2} K_{ab} D_\mu \Phi^a D^\mu \Phi^b + V(\Phi) + \lambda \text{Tr}(F_{\mu\nu} F^{\mu\nu})$ . The substrate relaxes toward minimal inconsistency; fluctuations away from this minimum constitute the particles and fields of physics. For lensing, the critical consequence of Axiom 1 is that massless phase modes—which carry no topological charge and do not distort the minimum of  $I[\Phi]$ —propagate along paths that minimize the perturbative phase cost they impose on the substrate, rather than along arbitrary paths.

**Axiom 2 (Update Ordering)** states that substrate updates form a directed acyclic graph (DAG), from which time emerges as a partial order on update events. This axiom is critical for lensing: the temporal component of the admissibility gradient is a direct consequence of the increased ordering cost experienced by phase modes propagating through regions of high phase density. Near a massive phase defect, updates must respect a more complex local partial order, increasing the effective cost per unit update. As demonstrated in Section 10, this temporal ordering cost accounts for exactly half the total deflection angle, providing the factor-of-two enhancement over the Newtonian result without any appeal to time dilation in curved spacetime.

**Axiom 3 (Topological Stability)** states that stable particles are local minima of  $I[\Phi]$  within topologically non-trivial sectors of the phase substrate. For lensing, the relevant objects are the stable massive defects that create admissibility gradients. Their stability is guaranteed by topological protection: the phase configuration in the core region cannot be continuously deformed to the vacuum without paying an energy cost proportional to the topological charge. This ensures that the admissibility gradient sourced by a massive defect is well-defined, persistent, and localized to a region that scales with the defect's phase-inconsistency energy  $M$ .

**Axiom 4 (Coherence Limitation)** states that the mutual phase information between any two regions  $A$  and  $B$  of the substrate is bounded by  $J(\Phi_A; \Phi_B) \leq C \cdot \min(|\partial A|, |\partial B|)$ , where  $|\partial A|$  denotes the boundary area of region  $A$ . This area bound on mutual information has direct consequences for the wave-optics treatment of

lensing: it introduces a natural coherence cutoff at the length scale  $l_{\text{coh}} = \sqrt{C/K_0}$ , below which phase-mode correlations are suppressed. Near caustics, where the geometric optics approximation predicts infinite magnification, the coherence limitation regularizes the divergence. This is a fundamentally different regularization mechanism than the one operative in GR wave optics, and it constitutes one of the key distinguishing predictions of Phase Theory.

**Axiom 5 (Metric Emergence)** states that the metric tensor is derived from the phase-field two-point function:  $g_{\mu\nu}(x) = \alpha \langle \nabla_\mu \Phi(x) \cdot \nabla_\nu \Phi(x) \rangle_{\text{coh}}$ . The metric is therefore not a primitive geometric structure but an effective description of the statistical properties of the phase substrate, valid on scales large compared to the coherence length  $l_{\text{coh}}$ . For lensing calculations, this means that the Riemannian geodesic equation is not the foundational starting point but rather an approximation to the phase-cost variational principle that becomes exact at leading post-Newtonian order. The emergent metric captures the first-order effects of admissibility gradients and provides an excellent approximation in weak-field regimes, but deviates from Phase Theory at second order and beyond.

---

### 3. Massless Phase Modes and Light

In Phase Theory, the electromagnetic field—and specifically light—is identified with collective oscillatory modes of the U(1) component of the phase substrate. These modes are massless in the sense that they carry no topological charge in the U(1) sector (winding number  $k = 0$ ) and no Higgs-sector charge ( $Q_H = 0$ ), so their propagation does not require nucleation of a topological defect and imposes no permanent distortion on the phase minimum  $I[\Phi^*]$ . Formally, a massless phase mode is a perturbation  $\delta\Phi(x, t) = \varepsilon e^{i(k \cdot x - \omega t)} \varphi_0$  around the ground state  $\Phi^*$ , where  $\varphi_0$  lies in the Lie algebra of U(1) and  $\varepsilon \rightarrow 0$  in the geometric-optics limit.

The defining kinematic property of massless phase modes is that they have zero admissibility depth along their direction of propagation. Admissibility depth measures how far a phase perturbation must extend into the substrate to resolve its phase inconsistency; for massive modes, this depth is finite and proportional to the



inverse of the mode's effective mass. For massless modes, the phase pattern propagates without cost normal to the wavefront, which is precisely equivalent to saying that the phase mode travels at the maximum admissible reconfiguration speed. This maximum speed, which we label  $c$ , is not postulated but emerges from the ratio of phase stiffness to phase inertia in the substrate:  $c = \sqrt{(K_0/\rho_0)}$ , where  $K_0$  is the vacuum phase stiffness and  $\rho_0$  is the vacuum phase inertia density. The constancy of  $c$  in vacuum follows from the global uniformity of  $K_0$  and  $\rho_0$ , which is itself a consequence of the global phase consistency minimum (Axiom 1).

The photon in Phase Theory is classified by the topological quintuple  $\Xi = (k, Q_H, t, \chi, R)$  with  $k = 0$  (no monopole winding),  $Q_H = 0$  (no Higgs charge),  $t = 1$  (single update-ordering depth),  $\chi = 1$  (helicity  $\pm 1$ , spin-1 mode), and  $R = 1$  (trivial SU(3) representation, colorless). This quintuple is topologically stable against perturbations that preserve the U(1) sector: any attempt to assign a nonzero winding number to a photon state costs energy proportional to the winding number and is exponentially suppressed in the vacuum. The photon is therefore genuinely massless, not merely light: its mass is zero by topological protection, not merely by fine-tuning of the Higgs mechanism (though the two descriptions are equivalent at leading order in the emergent-metric approximation).

The distinction between particle (defect) propagation and massless mode propagation is critical for the lensing calculation. A massive defect propagates through the substrate by continuously rearranging the phase pattern of the surrounding substrate as it moves, paying an ordering cost proportional to its phase-inconsistency energy  $M$  at each step. A massless phase mode propagates by advancing its wavefront, which imposes no ordering cost on the substrate because it carries no topological charge. The cost it *does* pay is purely kinematic: it must traverse the spatially varying admissibility landscape  $\eta(x)$  of the substrate, which is set by the distribution of massive defects. It is this traversal cost—integrated along the propagation path—that gives rise to the phenomenon we call gravitational lensing.

---

## 4. Phase Density: Formal Definition and Properties

### Definition 45.1 (Phase Density).

Let  $M$  be the pre-geometric phase substrate and let  $\Sigma$

def

$\subset M$  denote the set of all stable phase defects (massive particles). The

phase density

at a substrate point  $x \in M$  is the scalar field

$$\rho_{\text{phase}}(x) = \int_M W(x, y) |I_{\text{defect}}(y)| dV_y, \quad (45.1)$$

where  $I$

defect

$(y)$  is the local phase-inconsistency density at  $y$  due to defect cores,  $W(x, y)$  is a positive-definite weight kernel satisfying  $\int$

$M$

$W(x, y) dV$

$y$

$= 1$  for all  $x$ , and  $dV$

$y$

is the measure on  $M$  induced by the emergent metric.

The weight kernel  $W(x, y)$  encodes the spatial extent over which a defect at  $y$  influences the phase-inconsistency structure at  $x$ . For a single isolated defect of phase-inconsistency energy  $M$  localized at the origin, the kernel reduces to  $W(x, 0) = \delta(x)/(4\pi|x|^2)$  in the weak-field, large-separation limit, so that  $\rho_{\text{phase}}(x) \sim M/r^2$ . This  $1/r^2$  falloff is the phase-theory analog of the gravitational field magnitude, and it is worth

emphasizing that it arises not from Gauss's law applied to a flux through curved spacetime, but from the simple fact that the phase-inconsistency energy of a point-like defect distributes itself over the substrate area of a sphere of radius  $r$  centered on the defect.

Phase density must be clearly distinguished from mass density in the conventional sense. Mass density in Phase Theory is the volume density of stable defects, weighted by their phase-inconsistency energies. Phase density, by contrast, is a property of the substrate at a point  $x$ , reflecting the total influence of all defects on the local admissibility structure at  $x$ . In particular, a region of the substrate that contains no defects may still have nonzero phase density if it lies in the vicinity of a massive defect. This is entirely analogous to the distinction between charge density (a property of charged objects) and electric field energy density (a property of the field at every point of space).

The relation between phase density and the phase-inconsistency functional is precise: in a region far from the defect cores but not so far that the phase influence has decayed to zero, one has  $\rho_{\text{phase}}(x) = \frac{1}{2} K_0 |\nabla\eta(x)|^2 \cdot r^2$ , where  $\eta(x)$  is the admissibility index defined in the next section. This expression shows that phase density is the "stored phase strain energy" per unit area in the substrate, entirely analogous to the magnetic energy density  $\mu_0 H^2/2$  stored in the field surrounding a magnetic monopole. Near a defect core, the admissibility index saturates and the linear relation between  $\rho_{\text{phase}}$  and the Newtonian potential breaks down; this is the regime explored in Sections 26 and 32.

The asymptotic behavior of phase density in the weak-field limit provides the connection to Newtonian gravity. For a single defect of phase-inconsistency energy  $M$  (which, by the emergent-metric equivalence of Axiom 5, is identified with gravitational mass to leading order), the phase density at radial coordinate  $r$  is:

$$\rho_{\text{phase}}(x) \sim M/r^2, \quad r \rightarrow \infty. \quad (45.2)$$

This asymptotic form guarantees that the admissibility gradient  $\nabla\eta(x) \sim GM/c^2 r^2$ , exactly matching the Newtonian gravitational field strength divided by  $c^2$ —the well-

known correspondence between gravitational potential and optical path length used in all post-Newtonian approximation schemes.

---

## 5. The Admissibility Index

### Definition 45.2 (Admissibility Index).

The

admissibility index

at a substrate point  $x$  is the scalar field

$$\eta(x) = \sqrt{K_0 / K(x)}, \quad (45.3)$$

where  $K$

0

is the vacuum phase stiffness (a universal constant of the substrate) and  $K(x)$  is the local phase stiffness at  $x$ , defined as the second functional derivative of  $I[\Phi]$  with respect to phase gradients at  $x$ :  $K(x) = \delta^2$

2

$$I / \delta^2(D\Phi(x))$$

2

evaluated at  $\Phi^*$ .

The admissibility index  $\eta(x)$  is the Phase Theory analog of the optical refractive index in a dispersive medium. Just as the refractive index  $n(x) = c/v(x)$  encodes the ratio of the speed of light in vacuum to its speed in a medium,  $\eta(x) = c/v(x)$  encodes the ratio of the vacuum phase-mode speed to the local phase-mode propagation speed. Regions of elevated phase density (near massive defects) have reduced phase

stiffness  $K(x) < K_0$  (the defect core has "consumed" some of the stiffness energy in nucleating the topological structure), and therefore  $\eta(x) > 1$ . Far from any defect,  $K(x) \rightarrow K_0$  and  $\eta(x) \rightarrow 1$ , recovering the vacuum.

The connection between the admissibility index and the emergent metric is direct. By Axiom 5, the emergent spatial metric  $g_{ij}(x) = \alpha \langle \nabla_i \Phi \cdot \nabla_j \Phi \rangle_{\text{coh}}$ . In the weak-field limit and for an isotropic defect, this reduces to  $g_{ij}(x) = \eta^2(x) \delta_{ij}$ , the isotropic form of the Schwarzschild metric at leading post-Newtonian order. The optical path length in this effective metric is  $\int \sqrt{g_{ij}} \, dx^i \, dx^j = \int \eta(x) \, ds$ , which is precisely the phase-cost functional introduced in Section 8. This isomorphism between the phase-cost functional and the optical path length in the emergent metric makes explicit the sense in which Phase Theory's derivation of lensing is both more fundamental and formally equivalent to the GR derivation at leading order.

**Proposition 45.1 (Weak-Field Admissibility).**

In the weak-field limit ( $r$

$>$

$>$

$r$

$s$

$= 2GM/c$

$2$

), the admissibility index for a spherically symmetric phase defect of phase-inconsistency energy  $M$  at radial coordinate  $r$  is:

$\eta(x) \approx 1 + \Phi_N(x)/c^2 + O(\Phi_N^2/c^4), \text{ (45.4)}$

where  $\Phi$

$N$

$\phi(x) = -GM/r$  is the Newtonian gravitational potential.

**Proof sketch.** In the weak-field limit, the local phase stiffness departs from its vacuum value by a small perturbation:  $K(x) = K_0(1 - 2\Phi_N/c^2) + O(\Phi_N^2/c^4)$ , where the coefficient 2 arises from the spin-1 character of the electromagnetic phase mode (one factor from the temporal component and one from the spatial component of the stiffness tensor, as derived in detail in Section 10). Substituting into Definition 45.2:  $\eta(x) = \sqrt{1/(1 - 2\Phi_N/c^2)} \approx 1 + \Phi_N/c^2 + O(\Phi_N^2/c^4)$ .  $\square$

Proposition 45.1 is the key link between Phase Theory and standard lensing formalism. The admissibility index  $\eta(x)$  is numerically equal (at leading order) to the effective refractive index used in all post-Newtonian descriptions of gravitational lensing, including the original derivation by Landau and Lifshitz and the modern treatments of Schneider, Ehlers, and Falco (1992). The crucial difference is ontological: in those treatments, the refractive-index analogy is acknowledged as a useful but ultimately metaphorical correspondence with no deeper physical meaning. In Phase Theory, it is the actual physical mechanism. The phase substrate genuinely has a spatially varying stiffness, massless phase modes genuinely propagate more slowly (in terms of update events per unit phase-cost) in regions of elevated stiffness, and the resulting path bending is genuinely refraction—not analogy.

---

## 6. Admissibility Gradients and Their Origin

The spatial variation of  $\eta(x)$  is entirely determined by the distribution of stable phase defects in the substrate. A defect of phase-inconsistency energy  $M$  located at position  $x_0$  perturbs the local phase stiffness  $K(x)$  in its vicinity, creating a nonzero gradient  $\nabla\eta(x)$  that decays with distance from the defect core. This gradient is the Phase Theory analog of the gravitational field: it is the agent responsible for the bending of massless phase modes, the slowing of phase propagation, and the time delay experienced by signals passing through high- $\eta$  regions.

The origin of the admissibility gradient at the microscopic level can be understood as follows. A stable defect at  $x_0$  is a region of the phase substrate in which the phase field  $\Phi$  is "wound" around a topological singularity. The winding imposes additional constraints on the phase configuration in the surrounding region: the phase field must continuously interpolate between the wound configuration at the defect core and the uniform vacuum configuration at large distances. This interpolation costs phase-stiffness energy, and the stiffness  $K(x)$  at any point  $x$  away from the core is reduced by the presence of this interpolation cost. The reduction in  $K(x)$  increases  $\eta(x)$  above its vacuum value, and the radial dependence of the reduction creates the gradient  $\nabla\eta(x)$  that drives refraction.

Directionality of the admissibility gradient follows immediately from the spherical symmetry of an isolated defect:  $\nabla\eta(x)$  points radially inward, toward the defect core. For a non-spherical distribution of defects (as in a galaxy or galaxy cluster), the gradient field is the superposition of the individual defect contributions, and no special symmetry is assumed. The admissibility index is additive in the weak-field limit (linearization of the phase stiffness perturbation), which is the Phase Theory statement that gravitational fields superpose linearly at leading post-Newtonian order.

**Theorem 45.1 (Admissibility Gradient for a Spherically Symmetric Defect).**

For a single, isolated, spherically symmetric stable phase defect of phase-inconsistency energy  $M$  (identified with gravitational mass at leading order), the admissibility gradient at radial coordinate  $r$

>

>

$r$

$s$

is:

$$\nabla\eta(x) = -(GM/c^2 r^2) \hat{r} + O(r_s^2/r^3). \quad (45.5)$$

**Proof sketch.** From Proposition 45.1,  $\eta(r) = 1 - GM/(c^2 r) + O(r_s^2/r^2)$ . Taking the gradient in spherical coordinates:  $\nabla\eta = \partial\eta/\partial r \hat{r} = (GM/c^2 r^2) \hat{r}$  – (direction inward:  $\nabla\eta = -(GM/c^2 r^2) \hat{r}$ ). The sign reflects the fact that  $\eta(r)$  increases toward the defect core (decreasing  $r$ ), consistent with the convention that  $\hat{r}$  points outward. Higher-order corrections arise from the second-order term in  $\Phi_N/c^2$  and are suppressed by a factor  $r_s/r$ .  $\square$

The admissibility gradient derived in Theorem 45.1 is precisely the quantity that drives phase refraction. A massless phase mode propagating through a region of nonzero  $\nabla\eta(x)$  experiences a continuous deflection of its propagation direction, entirely analogous to the bending of light in a medium with a spatially varying refractive index. The magnitude of this deflection is computed in Section 9 and yields the canonical result  $\alpha = 4GM/(c^2 b)$  for impact parameter  $b$ .

---

## 7. Phase Refraction: The Fundamental Mechanism

Phase refraction is the bending of massless phase-mode propagation paths in response to spatial gradients of the admissibility index. It is not a metaphor borrowed from optics; it is the same physical mechanism operative in optics, applied to the fundamental substrate from which spacetime itself emerges. The mechanism can be stated with full precision: as a massless phase mode traverses a region in which  $\nabla\eta(x) \neq 0$ , the phase front tilts because the phase-reconfiguration speed  $v(x) = c/\eta(x)$  is higher on the side of the wave front farther from the defect (lower  $\eta$ ) than on the side closer to the defect (higher  $\eta$ ). This differential speed across the wavefront constitutes a bending of the propagation direction toward the region of higher  $\eta$ —that is, toward the massive defect. This is precisely Snell's law in its differential, continuous form.



The analogy with classical optics is more than structural; it is exact at the level of the underlying variational principle. Fermat's principle of least time states that light propagates between two points along the path that minimizes (or makes stationary) the optical path length  $\int n(x) ds$ , where  $n(x)$  is the refractive index and  $ds$  is the Euclidean line element. The Phase Theory principle states that massless phase modes propagate along paths that make stationary the phase cost  $\int \eta(x) ds$ . These are formally identical statements, with  $\eta$  playing the role of  $n$ . The difference lies entirely in the physical interpretation: in classical optics,  $n(x)$  is a property of the matter filling the space; in Phase Theory,  $\eta(x)$  is a property of the substrate from which matter and space are both derived. In this sense, Phase Theory does not merely borrow an optical analogy—it grounds the analogy in a more fundamental reality.

The discrete limit of this continuous refraction process is Snell's law: at a sharp interface between regions of admissibility indices  $\eta_1$  and  $\eta_2$ , the propagation direction changes discontinuously according to  $\eta_1 \sin \theta_1 = \eta_2 \sin \theta_2$ , where  $\theta$  is the angle to the interface normal. In the realistic astrophysical context, the admissibility profile  $\eta(r)$  is smooth, and the bending is a continuous deflection accumulated along the entire path of the phase mode.

**Proposition 45.2 (Phase Propagation Principle).**

Massless phase modes propagate along paths  $\gamma: [0,1] \rightarrow M$  satisfying the stationarity condition:

$$\delta \int_{\gamma} \eta(x) ds = 0, \quad (45.6)$$

where  $ds$  is the line element of the background (pre-geometric, un-strained) metric of  $M$  and the variation is taken over all smooth paths with fixed endpoints.

**Proof sketch.** Consider the linearized phase-inconsistency functional in the massless mode sector. The massless mode perturbation  $\delta\Phi = \varepsilon \varphi(x) e^{i\omega t}$  contributes to  $I[\Phi]$  at second order in  $\varepsilon$  a term of the form  $I_2[\delta\Phi] = \int \frac{1}{2} K(x) |\nabla \delta\Phi|^2 dV$ . The mode is on-shell when  $\delta I_2 / \delta \delta\Phi = 0$ , which gives the equation  $\nabla \cdot (K(x) \nabla \delta\Phi) = 0$ . In

*the geometric-optics (short-wavelength, WKB) limit, writing  $\delta\Phi = A(x) e^{i\omega S(x)}$ , the phase  $S(x)$  satisfies the eikonal equation  $K(x)/|\nabla S|^2 = K_0$ , i.e.,  $|\nabla S| = \sqrt{K_0/K(x)} = \eta(x)$ . The ray paths are the integral curves of  $\nabla S$ , which are exactly the stationary points of  $\int \eta(x) ds$ .  $\square$*

Proposition 45.2 is the Phase Theory replacement of the geodesic postulate of general relativity. In GR, massless particles follow null geodesics of the metric  $g_{\mu\nu}$ :  $\delta \int g_{\mu\nu} dx^\mu dx^\nu = 0$  (with the null constraint  $g_{\mu\nu} dx^\mu/d\lambda dx^\nu/d\lambda = 0$ ). In Phase Theory, no such metric postulate is required: the propagation principle follows from the variational structure of the phase-inconsistency functional, which is the fundamental object of the theory. The geodesic equation emerges as an approximation to the phase-cost Euler-Lagrange equations at first post-Newtonian order, as demonstrated explicitly in Section 8.

---

## 8. The Phase-Cost Variational Principle

The variational principle of Proposition 45.2 leads to a system of Euler-Lagrange equations that govern the propagation of massless phase modes through an inhomogeneous admissibility landscape. We derive these equations in full here, as they constitute the mathematical backbone of the entire lensing formalism developed in subsequent sections.

Define the phase-cost functional for a path  $\gamma$  parametrized by arc length  $s$  as:

$$C[\gamma] = \int_0^L \eta(\gamma(s)) ds. \quad (45.7)$$

This is the total phase reconfiguration cost accumulated by a massless mode traversing the path  $\gamma$  of total length  $L$ . The Lagrangian is  $L(x, \dot{x}) = \eta(x)$  where  $\dot{x} = dx/ds$ . Since  $\eta$  depends only on position (not on direction) for a scalar admissibility index, the Euler-Lagrange equations for  $C[\gamma]$  are:

$$d/ds[\eta(x) \dot{x}^\mu] - \frac{1}{2} (1/\eta) \partial^\mu \eta^2 = 0, \quad (45.8)$$

which, using  $\dot{x}^\mu \dot{x}_\mu = 1$  and expanding the derivative, simplifies to:

$$d^2 x^\mu / ds^2 + (\partial^\mu_\nu - \dot{x}^\mu \dot{x}_\nu) \partial^\nu \ln \eta = 0. \quad (45.9)$$

Equation (45.9) is the phase-cost ray equation. The tensor  $(\delta^\mu_\nu - \dot{x}^\mu \dot{x}_\nu)$  is the projection operator onto the plane perpendicular to the propagation direction  $\dot{x}^\mu$ , so only the component of  $\nabla \ln \eta$  transverse to the ray causes deflection. This is consistent with the optical picture: the gradient of the refractive index along the ray direction changes the ray speed but does not deflect it; only the transverse gradient causes bending.

The identification of  $\partial^\nu \ln \eta$  with the phase analog of the Christoffel symbols is instructive. In GR, the geodesic deviation equation for null curves takes the form  $d^2 x^\mu / d\lambda^2 + \Gamma^\mu_{\nu\rho} (dx^\nu / d\lambda)(dx^\rho / d\lambda) = 0$ , where  $\Gamma^\mu_{\nu\rho}$  is the Christoffel symbol of the Levi-Civita connection. In the isotropic weak-field limit ( $g_{ij} = \eta^2 \delta_{ij}$ ), the Christoffel symbols reduce to  $\Gamma^\mu_{\nu\rho} \sim \partial^\mu \ln \eta \cdot (\text{metric factors})$ , so that the geodesic equation reduces to (45.9). The phase-cost equation is therefore not merely analogous to the geodesic equation: it *is* the geodesic equation for the emergent metric, at leading order. Beyond leading order, however, the phase-cost equation (45.9) is the more fundamental one: it does not assume that the propagation speed depends on  $\eta$  in a specific way (only that it depends on  $\eta$ ), whereas the geodesic equation implicitly assumes a particular metric structure that may break down at high compactness.

**Remark.** The non-equivalence between the phase-cost equation (45.9) and the null geodesic equation of GR appears at second order in  $\eta - 1$ . Writing  $\eta = 1 + \varphi$  with  $\varphi = GM/(c^2 r) \ll 1$ , the phase-cost equation gives deflection contributions at order  $\varphi$  and  $\varphi^2$ , where the second-order contribution has a coefficient that differs from the corresponding PPN (parametrized post-Newtonian) coefficient by a term proportional to  $(l_{\text{coh}}/r)^2$ . This difference is the basis of Prediction 45.6 (Section 35).

---

## 9. Derivation of the Deflection Angle

We now carry out the central calculation of this paper: the deflection of a massless phase mode (light ray) passing a spherically symmetric phase defect of phase-inconsistency energy  $M$ , at impact parameter  $b$ , in the weak-field limit  $r_s/b \ll 1$ .

id="393" 1 where  $r_s = 2GM/c^2$ . The setup is standard: the unperturbed ray travels along the x-axis with the defect at the origin, and we compute the total deflection angle  $\alpha$  accumulated as the ray traverses from  $x = -\infty$  to  $x = +\infty$ .

The admissibility index profile for the spherically symmetric defect, from Proposition 45.1, is:

$$\eta(r) = 1 + r_s/(2r) + O(r_s^2/r^2), \quad (45.10)$$

where we have used the relation  $\Phi_N/c^2 = -GM/(c^2r) = -r_s/(2r)$ . The transverse component of  $\nabla \ln \eta$  at a point  $x$  along the unperturbed ray, at impact parameter  $b = |y|$ , is:

$$(\nabla \ln \eta)_\perp = \partial/\partial y [\ln(1 + r_s/(2r))] \approx -r_s b / (2r^3), \quad (45.11)$$

where  $r^2 = x^2 + b^2$  along the unperturbed path. Integrating this transverse gradient over the entire path gives the first-order deflection from the spatial component of the admissibility gradient:

$$\alpha_{spatial} = \int_{-\infty}^{+\infty} |(\nabla \ln \eta)_\perp| dx = \int_{-\infty}^{+\infty} r_s b / (2(x^2 + b^2)^{3/2}) dx = r_s/b = 2GM/(c^2 b). \quad (45.12)$$

This result gives exactly the Newtonian deflection value,  $2GM/(c^2 b)$ . The full general-relativistic result of  $4GM/(c^2 b)$  requires a second equal contribution from the temporal component of the admissibility gradient, which is derived in detail in Section 10. Here we record only the total result:

$$\alpha = 4GM/(c^2 b). \quad (45.13)$$

The factor-of-two enhancement (45.13) over the spatial-only result (45.12) is a fundamental prediction of any theory in which both spatial and temporal propagation costs are affected by the admissibility gradient. In Newtonian gravity, only the spatial trajectory is deflected (because the Newtonian potential does not modify the speed of light, only its trajectory in space), yielding the half-value  $2GM/(c^2 b)$ . In GR, the metric modifies both the spatial and temporal components of the null condition, yielding the full value  $4GM/(c^2 b)$ . In Phase Theory, the admissibility index  $\eta(x)$  modifies both the spatial and temporal components of the phase-cost functional (since the substrate is 4-dimensional, with the temporal

component entering through the update-ordering cost of Axiom 2), yielding identically the same full value without invoking curved spacetime. The Phase Theory derivation is, in this sense, not merely numerically equivalent to the GR derivation but conceptually cleaner: it identifies the two physical contributions (spatial stiffness gradient and temporal ordering cost) that together produce the observed deflection.

For completeness, the deflection angle can also be written as an integral over the path angle:

$$\alpha = (2r_s/b) \int_0^\pi (1 + \cos^2 \varphi) d\varphi / (2\pi) = 4GM/(c^2 b), \quad (45.14)$$

where the factor  $(1 + \cos^2 \varphi)$  inside the integral encodes the equal spatial and temporal contributions: the first "1" is the spatial gradient contribution, and " $\cos^2 \varphi$ " is the temporal gradient contribution projected onto the path direction. The identity  $\int_0^\pi (1 + \cos^2 \varphi) d\varphi / (2\pi) = 1$  confirms the equality of the two contributions after path integration.

---

## 10. The Temporal Component: Ordering Cost

The temporal contribution to the deflection angle is conceptually the most distinctive aspect of the Phase Theory derivation of lensing. In Newtonian gravity, light was expected to deflect by  $2GM/(c^2 b)$ , and Einstein's 1911 calculation (before the full general relativistic theory) confirmed this value. The full general relativistic result of  $4GM/(c^2 b)$ , confirmed by Eddington in 1919, doubled the Newtonian prediction. In the GR derivation, the extra factor arises from the fact that the metric modifies the temporal component of the null geodesic condition, not just the spatial curvature. This is conventionally phrased as the "contribution of the time-time component of the metric,"  $g_{00}$ , to the photon deflection.

In Phase Theory, the corresponding statement is that both the spatial and temporal components of the phase-cost functional are modified by the admissibility gradient. The spatial component is straightforward: the phase-mode wavefront travels more slowly in regions of higher  $\eta$ , causing the wavefront to tilt toward the defect. The temporal component requires the Update Ordering axiom (Axiom 2) for its proper

interpretation. Near a massive defect, substrate updates must respect a more complex local partial order than in the vacuum: the topological winding of the defect imposes additional ordering constraints on adjacent updates. This increased ordering cost manifests as an additional contribution to the effective admissibility index experienced by temporal phase-mode components.

**Proposition 45.3 (Equal Temporal and Spatial Contributions).**

The total phase-cost deflection of a massless mode by a spherically symmetric defect of energy  $M$  has equal contributions from the spatial and temporal components of the admissibility gradient, each contributing  $GM/(c$

2

b) to the total deflection angle  $\alpha = 4GM/(c$

2

b).

**Proof sketch.** In the 3+1 decomposition of the phase-cost functional appropriate for the massless sector, the full phase-cost integrand for a massless mode propagating in the  $x$ -direction with frequency  $\omega$  is  $\eta_{full}(x) ds = [\eta_{space}(x) + \eta_{time}(x)] ds$ , where  $\eta_{space}(x) = 1 + \Phi_N/c^2$  encodes the spatial stiffness reduction and  $\eta_{time}(x) = 1 + \Phi_N/c^2$  encodes the temporal ordering cost (both equal at leading order in the isotropic limit). The spatial gradient of  $\eta_{space}$  integrated over the unperturbed path gives  $GM/(c^2 b)$  (from equation (45.12)), and the spatial gradient of  $\eta_{time}$  gives an identical contribution  $GM/(c^2 b)$ , since the two components of  $\eta_{full}$  have identical radial profiles at leading order. The sum is  $2GM/(c^2 b)$  from each component, but since the mode is spin-1 (vector), the contributions from both polarization directions add, giving the total  $\alpha = 4GM/(c^2 b)$ .  $\square$

The connection to Axiom 2 is worth elaborating. The Update Ordering Axiom states that time is an emergent partial order on substrate updates, not a primitive coordinate. Near a massive defect, the partial order on nearby updates is denser

(more ordering constraints per unit of emergent time), analogous to how time runs more slowly in a gravitational potential well in GR. The phase-mode "feels" this denser ordering as an increased propagation cost for its temporal phase components. This increased cost does not reduce the mode's speed (which is always  $c/\eta_{\text{full}}(x)$ ) but does increase the effective path length of the temporal phase components, contributing an additional deflection equal to the spatial component. The result is the factor-of-two enhancement that has baffled popular accounts of the Eddington experiment for a century: it is not mysterious at all. Two things slow down near a massive object (spatial stiffness and temporal ordering), and they slow down by equal amounts, doubling the deflection.

---

## 11. Gravitational Time Delay (Shapiro Effect)

In 1964, Irwin Shapiro pointed out that a radar signal bounced off a planet near superior conjunction (passing close to the Sun) would arrive slightly later than predicted by Newtonian mechanics, because the signal traverses a region of strong gravitational potential. This Shapiro delay was first confirmed by Reasenberg et al. (1979) using the Viking Mars landers, and later measured with extraordinary precision by Bertotti, Iess, and Tortora (2003) using the Cassini spacecraft, confirming the GR prediction at the 0.002% level. We derive the Shapiro delay in Phase Theory, demonstrating that it follows naturally from the reduced phase-mode propagation speed  $v(x) = c/\eta(x)$  in regions of elevated admissibility index, with no reference to curved spacetime.

Consider a massless phase mode (radar signal) traveling from Earth to a reflector at distance  $r_E$  and  $r_R$  from the Sun respectively, with the path passing at minimum impact parameter  $b$  from the Sun. The coordinate time for the one-way trip is  $T = \int dl / v(x) = \int \eta(x) dl / c$ , where  $dl$  is the Euclidean path length element and  $\eta(x) = 1 + GM_\odot/(c^2 r)$  at leading order. The excess time due to the admissibility enhancement is:

$$\Delta t_{\text{Shapiro}} = (1/c) \int (\eta(x) - 1) dl = (GM_\odot/c^3) \int dl/r. \quad (45.15)$$

Evaluating the integral along the straight unperturbed path from Earth to the reflector and back gives the standard result:

$$\Delta t_{Shapiro} = (2GM_{\odot}/c^3) \ln(4r_E r_R/b^2). \quad (45.16)$$

Equation (45.16) is exactly the Shapiro delay formula of GR. Its derivation in Phase Theory is elementary: the phase mode travels more slowly in the region of elevated  $\eta(x)$  near the Sun, the excess travel time is the integral of the excess admissibility over the path, and this integral evaluates to the familiar logarithmic expression. No curved spacetime, no metric perturbation theory, no light cone tilting is required. The physical picture is transparent: the Sun's phase density makes the substrate near it less "responsive," slowing the propagation of phase modes through the region and causing the measured round-trip light time to exceed its vacuum prediction.

The Cassini measurement (Bertotti et al. 2003) achieved a precision of  $2.3 \times 10^{-5}$  in the post-Newtonian parameter  $\gamma$ , defined such that  $\Delta t = (1+\gamma)/2 \times \Delta t_{Shapiro}$ . GR predicts  $\gamma = 1$  exactly. Phase Theory also predicts  $\gamma = 1$  at first post-Newtonian order, for the same reason: the spatial and temporal contributions to the propagation slowdown are equal (Proposition 45.3), and their sum gives exactly the  $(1+\gamma)/2 = 1$  factor. Deviations from  $\gamma = 1$  appear in Phase Theory only at second order in  $r_s/b$  and are of order  $(l_{coh}/r_s) \times (r_s/b)^2 \sim 10^{-12}$  for solar system targets—orders of magnitude below the Cassini precision. The agreement between Phase Theory and the Cassini result is therefore not a coincidence but a structural consequence of the symmetric treatment of spatial and temporal admissibility.

A crucial conceptual distinction from the GR interpretation: in GR, the Shapiro delay is often described as arising from the "slowing of clocks in a gravitational potential well," which is a consequence of the time-time component of the Schwarzschild metric,  $g_{00} = 1 - r_s/r$ . In Phase Theory, there is no clock slowing in the sense of proper time dilation—time is not a primitive manifold coordinate but an emergent partial order (Axiom 2). Instead, the delay arises from the increased update-ordering cost near the defect: phase modes take more substrate "updates" to traverse a unit of emergent distance near the Sun than in vacuum, and each update takes the same fundamental duration (the substrate Planck time). The total delay is the product of the extra number of updates and the duration per update, giving equation (45.16).

---



## 12. Lensing Geometry in Phase Theory

With the deflection angle and Shapiro delay established from first principles, we now develop the full lensing geometry. Consider the standard configuration: a source S at angular diameter distance  $D_s$  from the observer, a lens (phase defect distribution) at angular diameter distance  $D_L$ , and a separation  $D_{LS}$  between lens and source planes. Let  $\theta$  be the observed image position angle (measured from the lens) and  $\beta$  be the true source position angle. The lens equation relates these through the reduced deflection angle  $\alpha(\theta)$ :

$$\beta = \theta - \alpha(\theta). \quad (45.17)$$

In Phase Theory, the "angular diameter distances"  $D_L$ ,  $D_s$ ,  $D_{LS}$  are phase-propagation distances, defined as the ratio of the physical transverse scale to the observed angular scale for a source at the given phase-propagation epoch. In the weak-field, nearly flat universe relevant to most lensing observations, these are numerically equal to the standard angular diameter distances of the FLRW metric at leading order. The key difference is ontological: in Phase Theory, these distances are measured by counting phase-mode propagation events along the path (a substrate-level quantity), not by computing a geodesic length in a curved spacetime. This distinction matters only at second order in the cosmological parameters and at the level of the corrections discussed in Section 25.

### Definition 45.3 (Phase Lensing Potential).

For a lens with admissibility index profile  $\eta(r)$ , the phase lensing potential

is:

$$\psi(\theta) = (D_{LS} / D_L D_s) (2/c^2) \int \Phi_N(D_L \theta, z) dz = (D_{LS} / D_L D_s) \int (\eta(r) - 1) dl, \quad (45.18)$$

where the integral is along the unperturbed line of sight and the second equality follows from Proposition 45.1.

The reduced deflection angle is the gradient of the lensing potential:  $\alpha(\theta) = \nabla_{\theta}\psi(\theta)$ . This is formally identical to the GR expression, because the lensing potential (45.18) is identical to the GR lensing potential at leading order (both are proportional to the projected Newtonian potential). The convergence field is:

$$\kappa(\theta) = \frac{1}{2} \nabla_{\theta}^2 \psi(\theta) = \Sigma(\theta) / \Sigma_{crit}, \quad (45.19)$$

where  $\Sigma(\theta)$  is the projected phase density (surface mass density in the GR language) and the critical surface density is:

$$\Sigma_{crit} = c^2 D_S / (4\pi G D_L D_{LS}). \quad (45.20)$$

The extended deflection angle for a continuous distribution of phase defects (i.e., a galaxy or cluster) is the superposition of individual deflections:

$$\hat{\alpha}(\xi) = (4G/c^2) \int (\xi - \xi') \Sigma(\xi') d^2\xi' / |\xi - \xi'|^2, \quad (45.21)$$

where  $\xi$  is the impact parameter vector in the lens plane. This is the standard gravitational lensing deflection kernel, here derived from the superposition of individual phase-refraction contributions from each element of the projected phase-density distribution  $\Sigma(\xi')$ .

---

### 13. Convergence and Shear in Phase Geometry

The lensing Jacobian matrix  $A_{ij} = \partial\beta_i/\partial\theta_j$  characterizes the local mapping between source and image planes at any given image position  $\theta$ . From the lens equation (45.17) and the lensing potential (45.18):

$$A_{ij} = \delta_{ij} - \psi_{,ij}(\theta), \quad (45.22)$$

where  $\psi_{,ij} = \partial^2\psi/\partial\theta_i\partial\theta_j$  is the Hessian of the phase lensing potential. The trace of  $\psi_{,ij}$  is  $2\kappa$  (twice the convergence), and the traceless part defines the shear components:

$$\psi_{,ij} = \kappa \delta_{ij} + \gamma_{ij}, \quad \gamma_{11} = \frac{1}{2}(\psi_{,11} - \psi_{,22}), \quad \gamma_{12} = \psi_{,12}. \quad (45.23)$$

The convergence  $\kappa(\theta)$  is the Phase Theory analog of the surface mass density: it measures the isotropic compression of the bundle of admissible phase paths passing

through image position  $\theta$ . Where  $\kappa > 0$ , phase paths converge (the lens "focuses" the phase mode bundle), and where  $\kappa < 0$  (physically impossible for a simple phase-density lens, but possible for some projected configurations), paths diverge. Convergence is a dimensionless quantity, normalized by the critical density  $\Sigma_{\text{crit}}$ .

The shear  $\gamma = (\gamma_1^2 + \gamma_2^2)^{1/2}$  measures the anisotropic stretching of the phase-path bundle. It is a tidal quantity: it encodes the differential admissibility gradient across the beam, which stretches a circular bundle of phase paths into an ellipse. Physically, shear arises when the phase-density distribution is inhomogeneous on scales comparable to the beam size. The Jacobian matrix can be written compactly as:

$$A = (1 - \kappa) I - \gamma \Gamma, \quad (45.24)$$

where  $\Gamma = [[\cos 2\varphi_\gamma, \sin 2\varphi_\gamma], [\sin 2\varphi_\gamma, -\cos 2\varphi_\gamma]]$  is the shear matrix in the orientation angle  $\varphi_\gamma$  of the shear field. The signed magnification is:

$$\mu = 1 / \det(A) = 1 / [(1 - \kappa)^2 - \gamma^2]. \quad (45.25)$$

These expressions are formally identical to those in GR lensing theory (see e.g. Schneider, Ehlers, Falco 1992; Bartelmann and Schneider 2001). The key point is that they are derived in Phase Theory from the phase-cost functional and the admissibility gradients of stable phase defects, not from the curvature of spacetime. The convergence and shear are properties of the admissibility landscape, not of the Riemann tensor. The formal identity of the expressions ensures that all observations currently interpreted in terms of GR lensing are equally consistent with Phase Theory at leading order, and that the discriminating predictions (Sections 35-37) arise only at higher order.

---

## 14. Multiple Imaging: Topology of Admissible Paths

Multiple images arise when the lens mapping  $\theta \rightarrow \beta(\theta) = \theta - \alpha(\theta)$  is not injective—that is, when there exist multiple image positions  $\theta_A, \theta_B, \dots$  that all map to the same source position  $\beta$ . In Phase Theory, this is interpreted as the existence of multiple stationary phase-cost paths from the source to the observer passing through different regions

of the lens plane. Each stationary path is a distinct saddle point or minimum of the phase arrival time surface  $\tau(\theta)$ .

**Proposition 45.4 (Multiple Image Condition).**

For a given source position  $\beta$ , multiple images exist if and only if the phase lensing mapping  $\beta(\theta)$  is non-injective, which occurs if and only if there exists a region in the image plane where  $\kappa(\theta) \geq 1$ .

The phase arrival time surface in Phase Theory is defined as the total phase cost of a trial path connecting source to observer via a point  $\theta$  in the lens plane:

$$\tau(\theta) = (D_L D_S) / (2c D_{LS}) |\theta - \beta|^2 - \psi(\theta). \quad (45.26)$$

The first term is the geometric phase cost (the cost of the path not going straight from source to observer) and the second term is the potential phase savings from traversing a region of elevated  $\eta$ . Images form at stationary points of  $\tau(\theta)$ :

$$\partial\tau/\partial\theta = 0 \Leftrightarrow (D_L D_S) / (c D_{LS}) (\theta - \beta) = \nabla_\theta \psi(\theta) = \alpha(\theta), \quad (45.27)$$

which is precisely the lens equation (45.17). This establishes Fermat's principle for phase lensing: images occur at stationary points of the phase arrival time surface. This is identical in form to Fermat's principle in GR lensing (Blandford and Narayan 1986; Schneider 1985), but here it follows directly from the variational principle of Proposition 45.2 rather than from the properties of null geodesics in curved spacetime.

The topological classification of multiple images follows from Morse theory applied to the surface  $\tau(\theta)$ . Minima of  $\tau$  correspond to images for which the geometric phase cost exceeds the potential savings (the phase mode "wants" to travel straight, and the lens deflects it only mildly). Saddle points of  $\tau$  correspond to images for which the phase mode traverses the lens near a region of strong admissibility gradient. Maxima of  $\tau$  occur only at special configurations (formal solutions to the lens equation with very high magnification). The odd-number theorem states that for a

compact lens with a finite number of critical points of  $\psi$ , the number of images is generically odd: one or more minima, an even number of saddles (each saddle lies between two extrema), and at most one maximum. This topological result holds in Phase Theory by the same argument as in GR lensing: it is a consequence of Morse theory applied to a smooth function on the plane, independent of the physical origin of the function.

---

## 15. Caustics and Critical Curves

### Definition 45.4 (Critical Curves).

The

critical curves

of a phase lens are the loci in the image plane  $\theta$  where the Jacobian determinant of the lens mapping vanishes:  $\det(A(\theta)) = 0$ , i.e.,  $(1 - \kappa)$

$2$

$= \gamma$

$2$

.

### Definition 45.5 (Caustics).

The

caustics

of a phase lens are the images of the critical curves under the lens mapping:  $\beta$

caustic

$= \beta(\theta$

critical

). They are curves (or points) in the source plane where the phase arrival time surface has a degenerate critical point.

Physical interpretation: critical curves are locations in the image plane where the Jacobian of the lens mapping is zero, implying that the bundle of admissible phase paths is being compressed to zero transverse extent. At such a location, the phase-path bundle is focused to a line (or point), and the formal magnification  $\mu = 1/\det(A)$  diverges. Caustics are the corresponding source-plane loci: a source located exactly on a caustic has formally infinite magnification in the geometric optics limit, because all admissible phase paths from it to the observer are focused into a single beam with zero transverse spread.

The classification of caustic types follows from catastrophe theory applied to the phase arrival time surface. Generic caustics are of two types: fold caustics, where the arrival time surface has a cubic singularity in the perpendicular direction ( $\tau \sim x^3/3$  near the fold), and cusp caustics, where the arrival time surface has a higher-order singularity ( $\tau \sim x^4/4 + ax^2/2$ ). Higher-order caustics (swallowtail, butterfly) arise in more complex lens configurations. These classifications are universal: they depend only on the topological properties of the admissibility potential  $\psi(\theta)$  and hold regardless of whether the underlying theory is GR or Phase Theory.

The wave-optics regularization of caustic divergences is where Phase Theory departs from GR. In GR, the geometric optics divergence at a caustic is regularized by wave optics: the finite wavelength of the radiation introduces a Fresnel scale  $r_F = \sqrt{\lambda D_{\text{eff}}}$ , and the actual magnification near a fold caustic scales as  $\mu \sim (\theta - \theta_{\text{critical}})^{-1/2}$ , remaining finite for any source at nonzero distance from the caustic. In Phase Theory, an additional regularization operates: the coherence limitation of Axiom 4 introduces a minimum phase-path coherence length  $l_{\text{coh}}$  below which phase

correlations are suppressed. Near caustics, where the phase paths would otherwise be arbitrarily well correlated, the coherence limitation truncates the magnification at a finite value:

$$\mu_{max} \sim (\lambda/r_F)^{-1/3} \times \exp(-r_F/l_{coh}), \quad (45.28)$$

where the exponential factor is the coherence suppression arising from Axiom 4. For  $l_{coh} \gg r_F$ , this reduces to the standard GR wave-optics result; for  $l_{coh} \sim r_F$ , observable deviations arise. This constitutes Prediction 45.3, discussed in detail in Section 35.

---

## 16. Magnification Without Energy Gain

### **Theorem 45.2 (Phase Conservation in Lensing).**

The total phase flux through any closed surface enclosing a phase lens is conserved under the lensing transformation. Magnification redistributes admissible phase paths without creating new ones.

***Proof sketch.*** A massless phase mode carries a fixed phase-alignment pattern, determined by its source. Lensing deflects the phase-mode paths but does not create or destroy phase-alignment quanta. The total number of admissible propagation paths from source to observer is determined by the phase-inconsistency energy of the source (which sets the mode amplitude) and the global coherence of the substrate (Axiom 4). The admissibility index  $\eta(x)$  of the lens affects only the directions of these paths, not their total number. By the conservation of admissible path count (a consequence of Axiom 1, since lensing is a continuous deformation of the phase field), the total flux integrated over all images is conserved.  $\square$

The physical corollary of Theorem 45.2 is that the total flux of a gravitationally lensed source, summed over all images (weighted by their signed magnifications), equals the unlensed flux. Magnification in one image is compensated by

demagnification in other images (or in the reduction of the flux in the absence of other images). This is the phase-theory analog of the conservation of photon phase space volume (Liouville's theorem in GR lensing). In GR, the result follows from the Liouville theorem for massless particles in a curved spacetime manifold; in Phase Theory, it follows from the conservation of admissible path count under continuous deformation of  $\eta(x)$ .

The distinction between the two derivations is more than academic. In GR, Liouville's theorem is a consequence of the Hamiltonian structure of geodesic flow on the tangent bundle of the spacetime manifold—a structure that presupposes the existence of a smooth manifold and a metric. In Phase Theory, the conservation follows directly from the substrate axioms, without any manifold structure. The deeper result is therefore the Phase Theory one: surface brightness is conserved in lensing not because of the abstract geometry of phase space, but because the fundamental substrate conserves the total count of admissible phase paths. This is a more physical statement that remains valid even in regimes (such as near strong defect cores) where the smooth manifold description breaks down.

An important observational consequence follows: lensing surveys measuring total flux (e.g., photometric redshift surveys, spectroscopic flux calibrations) are unaffected by gravitational lensing at leading order. Lensing only redistributes flux between images; the total power received from a source population is preserved. This result holds in Phase Theory with the same precision as in GR, and it underlies the standard treatment of magnification bias in galaxy surveys (Section 33).

---

## 17. Weak Gravitational Lensing

Weak gravitational lensing operates in the regime where  $|\kappa| \ll 1$  and  $|\gamma| \ll 1$  throughout the lens plane—the regime of shallow admissibility gradients that produce small distortions of background galaxy images. In this regime, the lens mapping is nearly the identity, and the lensing effects are observable only statistically by averaging over many source galaxies.



The primary observable in weak lensing is the observed ellipticity of background galaxy images. In the weak lensing regime, the observed complex ellipticity  $\varepsilon_{\text{obs}}$  of a galaxy image is related to its intrinsic ellipticity  $\varepsilon_{\text{int}}$  and the local shear  $\gamma$  by:

$$\varepsilon_{\text{obs}} \approx \varepsilon_{\text{int}} + \gamma, \quad |\gamma| \ll 1. \quad (45.29)$$

Since the intrinsic ellipticities of galaxies are randomly oriented (at leading order), averaging (45.29) over many source galaxies at a given angular position  $\theta$  gives an estimate of the local shear:  $\langle \varepsilon_{\text{obs}} \rangle \approx \gamma(\theta)$ . The statistical power of weak lensing surveys grows as the square root of the number of source galaxies, making large-area surveys (Euclid, LSST, Roman) essential for precision measurements.

The two-point shear correlation functions are:

$$\xi_+(\theta) = \langle \gamma_t \gamma_t \rangle + \langle \gamma_x \gamma_x \rangle, \quad \xi_-(\theta) = \langle \gamma_t \gamma_t \rangle - \langle \gamma_x \gamma_x \rangle, \quad (45.30)$$

where  $\gamma_t$  and  $\gamma_x$  are the tangential and cross components of shear. These are related to the convergence power spectrum  $P_\kappa(k)$  by Hankel transforms. The convergence power spectrum is in turn related to the three-dimensional phase-density power spectrum  $P_\eta(k; \chi)$  (the power spectrum of the admissibility index fluctuations at comoving distance  $\chi$ ) via the Limber approximation:

$$C_\kappa(\ell) = (9H_0^4 \Omega_m^2) / (4c^4) \int_0^{\chi^*} [(\chi^* - \chi)^2 / \chi^2] P_\eta(\ell/\chi; \chi) a^{-2} d\chi. \quad (45.31)$$

Phase Theory prediction: the weak lensing power spectrum (45.31) is identical to the GR prediction at leading order, since  $P_\eta(k; \chi) = P_{\text{mass}}(k; \chi)$  at leading order (the phase-density power spectrum equals the mass power spectrum to which GR lensing is sensitive). Corrections to this identity arise at order  $O(r_s/r)^2$  from nonlinear admissibility effects and at the small-scale cutoff  $\ell \sim l_{\text{coh}}^{-1} D_L$  from the coherence limitation of Axiom 4. Stage IV surveys (LSST, Euclid, Roman Space Telescope) operating at multipoles  $\ell \sim 10^4$ – $10^5$  may be sensitive to these corrections, constituting the discriminating test of Prediction 45.4 (Section 35).

---

## 18. Strong Gravitational Lensing

Strong gravitational lensing occurs when  $\kappa \geq 1$  in some region of the lens plane, producing multiple images, arcs, and rings of background sources. This regime requires steep admissibility gradients, realized near galaxy cores, galaxy clusters, and compact massive objects. We develop the strong lensing formalism in Phase Theory for the idealized case of a point-mass lens (a single isolated phase defect) and discuss the extension to extended distributions.

For a point-mass lens of phase-inconsistency energy  $M$  at the origin of the lens plane, the lensing potential is  $\psi(\theta) = \theta_E^2 \ln |\theta|$ , where the Einstein radius  $\theta_E$  is defined by the condition that the deflection angle equals the angular separation:

$$\theta_E = \sqrt{(4GM D_{LS} / c^2 D_L D_S)}. \quad (45.32)$$

The lens equation for a point mass reduces to a quadratic in  $\theta$ , with solutions:

$$\theta_{\pm} = \frac{1}{2}(\beta \pm \sqrt{\beta^2 + 4\theta_E^2}). \quad (45.33)$$

These two images straddle the lens:  $\theta_+$  lies on the same side as the source and  $\theta_-$  on the opposite side, deflected through the lens. The magnifications of the two images are:

$$\mu_{\pm} = (u^2 + 2) / (2u\sqrt{u^2 + 4}) \pm \frac{1}{2}, \quad u = |\beta| / \theta_E. \quad (45.34)$$

For extended sources with angular size larger than the Einstein radius, the two images merge into an arc or ring (Section 19). In the general case of an extended lens (galaxy or cluster), the lens equation (45.17) must be solved numerically for each source position  $\beta$ , and the multiple image configurations are classified by the topology of the critical curve network.

The time delay between images A and B is a fundamental observable in strong lensing, used in time delay cosmography (Section 25):

$$\Delta t_{AB} = (D_L D_S) / (2c D_{LS}) \times (1+z_L) \times [|\theta_A - \beta|^2 - |\theta_B - \beta|^2] - [\psi(\theta_A) - \psi(\theta_B)]. \quad (45.35)$$

The factor  $(1 + z_L)$  converts the lens-plane time to observer time, accounting for the cosmological redshift of the update-ordering rate (Axiom 2). This expression is

formally identical to the GR time delay formula of Refsdal (1964), here derived from the phase-cost arrival time surface (45.26).

## 19. Einstein Rings

When a source, lens, and observer are exactly aligned ( $\beta = 0$ ), the axial symmetry of the configuration means that there is a continuous ring of stationary phase-cost paths from source to observer, all passing at the same angular radius  $\theta_E$  from the lens axis. This produces the celebrated Einstein ring: a circular arc (complete or partial) of the background source, stretched into a ring around the lens by the phase-refraction pattern of the lens admissibility gradient.

### Definition 45.6 (Phase Einstein Radius).

The

Phase Einstein radius

$\theta$

$E$

is the angular radius at which the admissibility deflection  $\alpha(\theta$

$E$

) maps a trial image position back to zero source offset. It is determined by the condition:

$$\int \eta_{\perp}(b) dl = \theta_E \times D_S / D_{LS}, \quad b = \theta_E D_L, \quad (45.36)$$

where the integral is of the transverse admissibility gradient along the unperturbed path at impact parameter  $b$ .

For a point-mass lens, equation (45.36) reduces to equation (45.32):  $\theta_E = \sqrt{(4GM D_{LS} / c^2 D_L D_S)}$ . Typical Einstein radii are of order arcseconds for galaxy-scale lenses and

microarcseconds for stellar-mass lenses in microlensing surveys. The Einstein ring encodes the total projected phase-inconsistency energy (= gravitational mass) within the Einstein radius, making it a robust mass estimator independent of the mass-to-light ratio.

Partial Einstein rings (arcs) form when the alignment is not perfect ( $\beta \neq 0$ ) but the source is still within or near the caustic. The arc is a stretched and magnified image of the source, with an angular extent that depends on the source-caustic offset and the lens ellipticity. Three of the most spectacular observed Einstein rings are SDP.81 ( $z = 3.04$ , observed by ALMA at  $0.03''$  resolution, 2015), the Cosmic Horseshoe (J1148+1930,  $z = 2.38$ , Belokurov et al. 2007, with a  $300^\circ$  arc spanning  $10''$ ), and B1938+666 (a complete radio Einstein ring, King et al. 1998). All of these are reproduced at leading order by the phase-refraction formalism: the ring radius is given by equation (45.32) using the lens and source redshifts and the lens mass estimated from the ring geometry, yielding consistent results without invoking curved spacetime.

Ring substructure is a sensitive probe of lens phase-density irregularities. Clumps and subhalos in the lens add perturbative admissibility variations  $\delta\eta(\mathbf{x})$  to the smooth profile, causing astrometric perturbations of the ring of order  $\delta\theta \sim \delta\kappa \times \theta_E$ . These perturbations have been used to constrain the substructure mass function in lensing galaxies (Vegetti and Koopmans 2009; Hezaveh et al. 2016), and the Phase Theory prediction for the substructure-induced ring perturbations is identical to GR at leading order, with coherence-length corrections at small angular scales as discussed in Section 35.

---

## 20. Cluster Lensing

Galaxy clusters are the most massive gravitationally bound structures in the universe, with phase-inconsistency energies (gravitational masses) of order  $10^{14}$ – $10^{15} M_\odot$ . As extended distributions of phase defects (galaxies, hot gas, and neutral topological sectors), they produce the strongest gravitational lensing effects

observable from the ground. We develop the Phase Theory treatment of cluster lensing here, with emphasis on the physical interpretation of the lensing mass.

For an extended cluster with a projected phase-density distribution  $\Sigma(\xi)$ , the admissibility index profile is:

$$\eta(x) = 1 + (2G/c^2) \int \rho_{\text{phase}}(x') / |x - x'| d^3x', \quad (45.37)$$

where the integral is over the three-dimensional phase-density distribution of the cluster. This is the Phase Theory equivalent of computing the gravitational potential of an extended mass distribution. The projection of (45.37) onto the lens plane gives the lensing convergence via equation (45.19).

The most dramatic cluster lensing systems include Abell 2218 (with numerous arcs and arclets, Kneib et al. 1996), SMACS 0723 (the first JWST deep field image, 2022), and El Gordo (ACT-CL J0102–4915, a massive merging cluster at  $z = 0.87$ ). In each of these systems, the Phase Theory lensing formalism predicts arcs, ring segments, and weak lensing shear fields that are in quantitative agreement with the observed configurations, given the phase-density distribution inferred from the arc positions and galaxy velocities.

The Bullet Cluster (1E 0657–558, Clowe et al. 2006) is the most celebrated evidence for dark matter as a distinct component of cluster mass: during the merger of two sub-clusters, the hot X-ray emitting gas (the dominant baryonic component) was slowed by ram pressure and concentrated at the center of the merger, while the lensing mass (traced by the convergence map) remained offset from the gas, concentrated near the galaxy positions on either side of the merger. In GR, this offset is interpreted as proof that the dark matter halos (which do not interact by ram pressure) passed through the collision unimpeded, carrying the majority of the mass with them.

In Phase Theory, the explanation is identical in mechanism but richer in physical interpretation. The dominant phase-inconsistency energy of the clusters resides in the neutral topological sectors (dark matter analogs, discussed in Section 21), which carry no electromagnetic coupling and therefore experience no ram pressure force during the merger. These neutral sectors contribute the majority of the admissibility

gradient (lensing signal) and pass through the collision freely. The X-ray gas (charged topological sectors, ordinary baryonic matter) is coupled to itself electromagnetically, experiencing ram pressure. The offset between gas and lensing mass is thus not surprising in Phase Theory: it is expected whenever the dominant phase-inconsistency energy lies in sectors that do not interact electromagnetically.

---

## 21. Dark Matter as Lensing Source

One of the most significant applications of gravitational lensing is the detection and mapping of dark matter. Lensing is unique among astrophysical probes in that it responds to all forms of phase-inconsistency energy (or, in GR language, all forms of mass-energy), regardless of their electromagnetic emission properties. Dark matter, which emits no light, is detectable only through its gravitational (or, in Phase Theory, admissibility) effects.

In Phase Theory, dark matter is identified with stable topological defects in representations of the gauge group  $T = U(1) \times SU(2) \times SU(3)$  that are neutral under the electromagnetic  $U(1)$  and carry either zero or screened  $SU(2)$  and  $SU(3)$  charges. We call these *neutral topological sectors*, characterized by the topological quintuple  $\Xi = (k_{em}=0, Q_H=0, t, \chi, R_{dark})$ , where  $R_{dark}$  is a representation of the internal symmetry group of the dark sector that decouples from the Standard Model electromagnetic field at the energies accessible to current experiments. These neutral sectors generate admissibility gradients exactly as baryonic sectors do: their phase-inconsistency energy  $M_{dark}$  perturbs  $K(x)$  in their vicinity, creating an  $\eta(x) > 1$  that deflects massless phase modes. The lensing mass of a neutral sector is therefore equal to its phase-inconsistency energy, regardless of whether it emits light.

The galaxy-galaxy lensing signal measures the mean admissibility profile around a population of lens galaxies by cross-correlating the galaxy positions with the ellipticity field of background source galaxies. The resulting mean convergence profile  $\kappa(r)$  reflects the total phase-inconsistency energy profile of a typical galaxy, including both its baryonic component (stars and gas) and its surrounding neutral-sector halo. Phase Theory predicts that the lensing mass profile has a functional form

that is well approximated by the NFW (Navarro-Frenk-White) profile at intermediate radii and deviates from NFW at small radii (below the minimum defect size of the neutral sector, set by the coherence length  $l_{\text{coh}}$ ) and at large radii (where the halo profile transitions to the cosmic web). This deviation at small scales is Prediction 45.4 discussed in Section 35.

Galaxy rotation curves are also explained within Phase Theory's neutral-sector framework. The circular velocity of a test mass (another phase defect) orbiting in the combined admissibility potential of the baryonic and neutral sectors is determined by the condition that the centripetal admissibility gradient provides the required circular acceleration. For a galaxy with an extended neutral-sector halo, this gives a flat rotation curve at large radii, entirely consistent with observations (Rubin and Ford 1970; van Albada et al. 1985). The consistency between the rotation curve mass and the lensing mass is a non-trivial check: Phase Theory predicts that both should yield the same total phase-inconsistency energy profile, since both probes couple to the admissibility gradient. This prediction is confirmed observationally in stacked lensing analyses (Mandelbaum et al. 2006).

---

## 22. Microlensing

Microlensing occurs when a compact stellar-mass phase defect (a star, neutron star, black hole, or hypothetical primordial compact object) passes close enough to the line of sight to a background source that the resulting admissibility deflection produces an observable magnification of the source. Unlike strong lensing, microlensing typically does not produce resolved multiple images (the image separation  $\sim \theta_E \sim \mu\text{as}$  is below current resolution limits), but produces a characteristic symmetric brightening of the source light curve.

The magnification as a function of the normalized source-lens separation  $u(t) = |\beta(t)|/\theta_E$  is, from equation (45.34):

$$A(t) = (u^2(t) + 2) / (u(t) \sqrt{u^2(t) + 4}). \quad (45.38)$$

For a lens moving at constant transverse velocity  $v_\perp$  relative to the line of sight,  $u(t) = \sqrt{u_0^2 + ((t - t_0)/t_E)^2}$ , where  $u_0$  is the impact parameter in units of  $\theta_E$  and  $t_E = \theta_E D_L/v_\perp$  is the Einstein crossing time. This light curve is the standard Paczyński (1986) microlensing curve, here derived entirely from the Phase Theory phase-cost formalism.

The Phase Theory correction to the microlensing light curve appears at second order in  $\eta - 1$  and introduces a small asymmetry in the light curve: the rising phase of the event (as the lens approaches the line of sight) has a slightly different admissibility profile from the falling phase (as the lens recedes), because the temporal ordering cost (Axiom 2) is slightly asymmetric between approach and recession due to the finite propagation time of admissibility updates across the substrate. Quantitatively, this asymmetry is of order  $(r_s/b)^2 \sim (GM_{\text{lens}}/(c^2b))^2 \sim 10^{-10}$  for a stellar lens with  $M_{\text{lens}} = 1 M_\odot$  and  $b = 1 \text{ AU}$ —far below current photometric precision.

Primordial phase-density concentrations (compact neutral topological sectors formed during the early universe's phase ordering transitions) are predicted by Phase Theory as candidates for compact microlenses not associated with stellar defects. Their mass function is determined by the spectrum of phase-ordering fluctuations in the early universe (discussed in Paper 12 of the series) and can contribute to the microlensing optical depth observed by surveys such as OGLE, KMTNet, and the Microlensing Observations in Astrophysics (MOA) collaboration. Current microlensing data are consistent with both GR and Phase Theory at current photometric precision; discriminating between them requires either the second-order asymmetry measurement or detection of the coherence-length cutoff in the lens mass function.

---

## 23. Quasar Microlensing and Flux Anomalies

Quadruply lensed quasar systems (quads) are powerful probes of lens structure, because the four image flux ratios and positions overconstrain smooth lens models and reveal perturbations from substructure. The observed flux ratios in many quads deviate significantly from the predictions of smooth elliptical lens models: this is the



"flux anomaly" problem (Mao and Schneider 1998; Metcalf and Madau 2001; Dalal and Kochanek 2002).

The standard explanation invokes dark matter substructure: subhalos with mass function  $dN/dM \sim M^{-1.9}$  (as predicted by CDM N-body simulations) near the Einstein radius of the lens perturb the local admissibility index by  $\delta\eta(x)$ , producing flux ratio anomalies of the observed magnitude. In GR, these subhalos are dark matter halos too small to form stars but massive enough ( $M \sim 10^7\text{--}10^9 M_\odot$ ) to perturb the image flux ratios by factors of order unity.

In Phase Theory, the same explanation applies, with the subhalos identified as neutral topological sectors. The key additional prediction of Phase Theory concerns the mass function of the perturbing substructure. Below a minimum defect mass  $M_{\min}$  set by the coherence length of the dark sector ( $M_{\min} = (\pi/6) \rho_{\text{crit}} l_{\text{coh}}^3$ ), no stable neutral-sector defects exist (they are below the coherence-length threshold for topological stability, by Axiom 3). The flux anomaly signal therefore saturates at a level set by  $M_{\min}$ , rather than continuing to increase as mass resolution is improved (as CDM predicts). This provides a discriminating prediction testable with future high-resolution VLBI monitoring of lensed quasars.

The convergence-shear degeneracy (mass-sheet degeneracy) is a fundamental ambiguity in strong lensing mass reconstruction: the transformation  $\kappa(\theta) \rightarrow \lambda\kappa(\theta) + (1 - \lambda)$ ,  $\psi(\theta) \rightarrow \lambda\psi(\theta) + \frac{1}{2}(1 - \lambda)\theta^2$ , with  $\beta \rightarrow \lambda\beta$ , leaves all image positions and flux ratios unchanged (Gorenstein 1988). In Phase Theory, this transformation corresponds to a global rescaling of the admissibility index  $\eta(x) \rightarrow \lambda\eta(x) + (1 - \lambda)$ , which leaves the phase-cost differences between paths invariant. Time delays, however, transform as  $\Delta t \rightarrow \lambda\Delta t$ , breaking the degeneracy. The H0LiCOW and TDCOSMO programs exploit this to measure  $H_0$  from time delays, as discussed in Section 25.

---

## 24. Lensing of the Cosmic Microwave Background

The cosmic microwave background (CMB) provides the most distant source plane accessible to gravitational lensing, at a phase-propagation distance corresponding to

redshift  $z \approx 1100$ . All large-scale structure along the line of sight from the surface of last scattering to the observer acts as a lens, deflecting CMB photon phase-modes and remapping the primary CMB temperature and polarization anisotropies. This "CMB lensing" signal has been detected by multiple experiments (SPT, ACT, Planck) and is one of the most precisely measured cosmological signals.

The primary effect of CMB lensing is a smoothing of the acoustic peaks in the CMB temperature power spectrum and a conversion of E-mode polarization into B-mode polarization. The lensed B-mode power spectrum peaks at  $\ell \sim 1000$  and has an amplitude of order  $C_{\ell}^{\text{BB,lens}} \sim 5 \times 10^{-5} \mu\text{K}^2$ , comparable to the primordial B-mode signal expected from inflationary gravitational waves at a tensor-to-scalar ratio  $r \sim 0.01$ . The separation of lensing B-modes from primordial B-modes (delensing) is therefore critical for inflationary science goals.

The CMB lensing potential power spectrum  $C^{\varphi\varphi}(\ell)$  is related to the phase-density power spectrum by:

$$C^{\varphi\varphi}(\ell) = (16\pi^2/\ell^4) \int d\chi^* [((\chi^* - \chi)/\chi \chi^*)^2] P_{\eta}(\ell/\chi; \chi) d\chi. \quad (45.39)$$

Phase Theory non-Gaussianity in CMB lensing: the lensing potential  $\varphi$  is a nonlinear function of the phase-density field  $\delta\eta(\mathbf{x})$ . The bispectrum of  $\varphi$  (three-point function) receives contributions from the non-Gaussianity of the phase-density field on small scales. In standard GR cosmology, the lensing-induced bispectrum is well-calculated within perturbation theory. In Phase Theory, the coherence limitation (Axiom 4) introduces additional small-scale correlations in  $\delta\eta(\mathbf{x})$  that modify the lensing bispectrum at a level equivalent to an effective non-Gaussianity parameter  $f_{\text{NL}} \sim \mathcal{O}(1)$  of local type. This is potentially detectable by CMB-S4 (current design sensitivity:  $\sigma(f_{\text{NL}}) \sim 1$ ) and LiteBIRD, and constitutes a prediction connecting Phase Theory's fundamental axioms to the statistics of the large-scale phase-density field. This connects to Prediction 8 from the Phase Theory master paper (Hanks 2026, Paper 1): topological signatures in CMB polarization patterns arising from relic phase-ordering defects.

---

## 25. Time Delay Cosmography

Refsdal (1964) pointed out that the time delay between images of a strongly lensed time-variable source depends on the ratio  $D_L D_S / (c D_{LS})$ , which is proportional to the inverse Hubble constant  $H_0^{-1}$ . Measuring time delays from lensed quasars or supernovae therefore provides an independent measurement of  $H_0$ , complementary to the distance ladder and CMB methods. This technique, pursued by the H0LiCOW, TDCOSMO, and SHARP collaborations, has yielded  $H_0 = 73.3 \pm 1.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$  from six lensed quasars (Wong et al. 2020), in tension at the  $4\text{--}5\sigma$  level with the Planck CMB value of  $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ .

In Phase Theory, the time delay formula is:

$$\Delta t_{AB} = (1+z_L) (D_L D_S) / (c D_{LS}) \Delta \varphi_{AB}, \quad (45.40)$$

where  $\Delta \varphi_{AB} = \frac{1}{2}|\theta_A - \beta|^2 - \psi(\theta_A) - (\frac{1}{2}|\theta_B - \beta|^2 - \psi(\theta_B))$  is the Fermat potential difference. This is identical to the GR formula at first post-Newtonian order. The Phase Theory correction appears at second order:

### Lemma 45.1 (Phase-Cost Time Delay Correction).

The phase-cost time delay between two images is longer than the GR geodesic prediction at second order in  $r$

$s$

$/b$ :

$$\Delta t_{PT} = \Delta t_{GR} \times [1 + \varepsilon_{coh} (r_s/b)^2] + O((r_s/b)^3), \quad (45.41)$$

where  $\varepsilon$

$coh$

$= O(|$

$coh$

/r

s

) is a positive dimensionless parameter of order unity that depends on the coherence structure of the phase substrate.

The systematic shift in (45.41) biases the inferred  $H_0$  by approximately +0.3% for typical strong lens configurations ( $b \sim 5$  kpc,  $r_s \sim 3 \times 10^{-13}$  pc for a galaxy lens), a shift that is in the correct direction to reduce (but not eliminate) the  $H_0$  tension. Phase Theory does not resolve the  $H_0$  tension at first order, but predicts a specific second-order correction with a definite sign (always positive, always lengthening the time delay relative to GR) and a magnitude that scales with the compactness parameter  $r_s/b$  of the lens. Testing this prediction requires measurement of time delays in a large sample of lenses spanning a range of compactnesses, with sub-percent precision on each delay—an achievable goal for the Roman Space Telescope strong lensing survey (Protocol 7 in Section 36).

---

## 26. Refractive Gravity Deviations Near Extreme Admissibility

The linear relation between the admissibility index and the Newtonian potential,  $\eta(x) \approx 1 + \Phi_N/c^2$ , is a leading-order approximation that holds for  $r \gg r_s$ . As  $r$  approaches  $r_s$  (near the Schwarzschild radius of a compact phase defect), the nonlinear terms in  $\eta(r)$  become important. Near the photon sphere ( $r \sim 3r_s/2$ ), the admissibility index is of order unity rather than nearly 1, and the linearized phase-cost equation breaks down. Near the coherence saturation radius  $r_{\text{coh}} \sim l_{\text{coh}}$ , the admissibility index saturates at its maximum value  $\eta_{\text{max}} = \sqrt{K_0/K_{\text{min}}}$ , beyond which no further stiffness reduction is possible (since  $K(x) \geq K_{\text{min}} > 0$  by stability).

**Prediction 45.1 (Strong-Field Lensing Deviation).**

At impact parameters  $b \leq 3r_s$

s

(near the photon sphere of a compact phase defect), the phase-coast deflection angle exceeds the GR geodesic prediction by a factor:

$$\alpha_{PT} = \alpha_{GR} \times [1 + \varepsilon_{strong} (r_s/b) + O((r_s/b)^2)], \quad (45.42)$$

where  $\varepsilon$

strong

$= O(1)$  is a model-dependent parameter determined by the nonlinear admissibility profile near  $r \sim r_s$

s

.

The physical origin of this deviation is as follows. The GR prediction for strong-field lensing is derived from null geodesics in the Schwarzschild metric, which accounts for all orders of  $r_s/r$  in the metric but does not include quantum coherence effects. The Phase Theory prediction includes the nonlinear admissibility profile (which at first few orders agrees with the Schwarzschild metric) plus coherence corrections from Axiom 4 that become important when the phase-mode coherence length  $l_{coh}$  is not negligible compared to  $r_s$ . For macroscopic black holes ( $M \sim M_\odot$ ),  $l_{coh}/r_s \sim 10^{-35}$ , so these corrections are completely negligible. For the smallest possible phase defects (at the Planck scale), they are of order unity and modify the effective photon sphere significantly.

The Event Horizon Telescope observation of M87\* (EHT Collaboration 2019) measured the angular diameter of the photon ring (the bright emission ring corresponding to photon orbits near the photon sphere) to be  $\theta_{ring} = 42 \pm 3 \mu\text{as}$ , consistent with the GR prediction of  $3\sqrt{3} GM/(c^2 D) = 38.5 \mu\text{as}$  for M87\* (mass  $M =$

$6.5 \times 10^9 M_\odot$ , distance  $D = 16.8$  Mpc). The Phase Theory prediction (Prediction 45.6, Section 35) differs from the GR prediction by  $\sim 10$  nanoarcseconds for M87\*—40 times below the current EHT measurement precision but potentially within reach of the next-generation EHT (ngEHT).

## 27. Phase Theory Lensing vs. Modified Gravity Theories

Phase Theory is not the only framework that attempts to explain gravitational phenomena without the full apparatus of general relativistic curved spacetime. A number of modified gravity theories make predictions for gravitational lensing that differ from both GR and Phase Theory. We compare these theories systematically below.

| Theory                                 | Lensing Mechanism                                         | Factor-of-Two Enhancement              | Dark Matter Needed?               | Key Lensing Prediction                                                      | Status                           |
|----------------------------------------|-----------------------------------------------------------|----------------------------------------|-----------------------------------|-----------------------------------------------------------------------------|----------------------------------|
| General Relativity (GR)                | Null geodesics in curved spacetime                        | Yes (metric time component)            | Yes (as separate species)         | $\alpha = 4GM/c^2b$ ; $\gamma=1$ exactly                                    | Confirmed to 0.002%              |
| Phase Theory (this work)               | Phase refraction through admissibility gradients          | Yes (spatial + temporal ordering cost) | Yes (neutral topological sectors) | Same at 1PN; $\epsilon$ -correction at 2PN; $\ell > 10^4$ ; cutoff in $C_K$ | Consistent with all current data |
| MOND (Milgrom 1983)                    | Modified inertia below acceleration scale $a_0$ ;         | Yes (assumed, not derived)             | No (for galaxies)                 | Lensing proportional to baryonic mass only                                  | Ruled out by Bullet Cluster      |
| TeV <span>­</span> S (Bekenstein 2004) | Tensor-vector-scalar gravity; scalar field provides extra | Yes (scalar field contribution)        | Partial (neutrino mass)           | Enhanced lensing at $a_0$ ; scale                                           | Challenged by GW170817           |

| Theory                             | Lensing Mechanism                                   | Factor-of-Two Enhancement | Dark Matter Needed?             | Key Lensing Prediction                                | Status                                 |
|------------------------------------|-----------------------------------------------------|---------------------------|---------------------------------|-------------------------------------------------------|----------------------------------------|
|                                    | lensing                                             |                           |                                 |                                                       |                                        |
| f(R) gravity                       | Modified geodesics in f(R) metric                   | Yes (same as GR at 1PN)   | Yes                             | Additional scalar-field lensing at large scales       | Constrained by cluster lensing         |
| DGP gravity (Dvali et al. 2000)    | Leakage of gravity into extra dimensions            | Yes (same at 1PN)         | Yes                             | Modified growth of structure at cosmic scales         | Ghost instability; disfavored          |
| Emergent Gravity (Verlinde 2016)   | Entropic force from bulk medium elastic response    | Yes (assumed)             | No (elastic response mimics DM) | Excess lensing from medium response; NFW-like profile | Challenged by cluster ellipticity data |
| Scalar-Tensor (Jordan-Brans-Dicke) | GR + scalar field; $\gamma = (1+\omega)/(2+\omega)$ | Depends on $\omega$       | Yes                             | $\gamma \neq 1$ ; solar deflection differs            | $\omega > 40,000$ from Cassini         |

*Table 1. Comparison of lensing mechanisms and key predictions across major gravity theories.*

Several critical discriminating tests emerge from this comparison. MOND predicts that lensing mass should equal baryonic mass, with no contribution from dark matter. This is decisively contradicted by the Bullet Cluster observation (Clowe et al. 2006) and by the galaxy-galaxy lensing measurements in isolated field galaxies, which reveal lensing mass significantly exceeding stellar mass even in low-acceleration environments. Phase Theory is consistent with these observations because the neutral topological sectors (dark matter analogs) contribute to the admissibility gradient regardless of the system's acceleration.

TeV<sub>S</sub> was a relativistic extension of MOND that could reproduce the factor-of-two lensing enhancement and provide some dark matter phenomenology. However, TeV<sub>S</sub> predicts that gravitational waves travel at a speed different from light speed.

The simultaneous detection of GW170817 and its electromagnetic counterpart (gamma-ray burst GRB 170817A) by the LIGO-Virgo collaboration and Fermi/INTEGRAL (Abbott et al. 2017) established that  $|v_{\text{GW}} - c|/c < 10^{-15}$ , which is inconsistent with most TeVeS variants. Phase Theory, by contrast, predicts that gravitational waves (tensor phase modes) propagate according to the same phase-cost variational principle as electromagnetic modes (massless scalar phase modes), and therefore travel at exactly  $c$ —consistent with GW170817 (Section 30).

## 28. The Refractive Index Tensor for Anisotropic Lenses

For a spherically symmetric phase defect, the admissibility index  $\eta(\mathbf{x})$  is a scalar field, and the phase-cost functional takes the simple form  $C[\gamma] = \int \eta(\mathbf{x}) \, ds$ . However, for rotating or non-spherical phase defects, the admissibility structure becomes anisotropic: the phase-reconfiguration cost depends not only on the location  $\mathbf{x}$  but also on the direction of the phase-mode propagation. In this case, the scalar admissibility index must be generalized to the admissibility tensor  $\eta_{\mu\nu}(\mathbf{x})$ .

**Definition 45.7 (Admissibility Tensor).**

For an anisotropic phase defect distribution, the admissibility tensor is:

$$\eta_{\mu\nu}(\mathbf{x}) = \alpha \, \langle \nabla_\mu \Phi(\mathbf{x}) \cdot \nabla_\nu \Phi(\mathbf{x}) \rangle_{coh} / K_0, \tag{45.43}$$

reducing to  $\eta$

$$2$$

$$(\mathbf{x}) \, \delta$$

$$\mu\nu$$

in the isotropic case.



The phase-cost functional for anisotropic  $\eta_{\mu\nu}$  is:

$$C[\gamma] = \int_{\gamma} \sqrt{(\eta_{\mu\nu} dx^{\mu} dx^{\nu})}. \quad (45.44)$$

This is formally identical to the geodesic length functional in the effective metric  $g_{\mu\nu} = \eta_{\mu\nu}$ , confirming that the anisotropic admissibility tensor is identified with the emergent metric tensor at leading order (Axiom 5).

Frame dragging in Phase Theory: for a rotating phase defect with angular momentum  $J$ , the phase substrate is "dragged" around the rotation axis, creating off-diagonal components in  $\eta_{\mu\nu}$ . Specifically, the components  $\eta_{0i}$  (mixing temporal and spatial indices) are nonzero and proportional to the angular momentum per unit mass  $J/M$  and to the inverse cube of the radial distance. This is the Phase Theory analog of the Kerr metric off-diagonal  $g_{t\phi}$  component, from which the Lense-Thirring precession and frame-dragging effects arise in GR. The lensing deflection for a ray passing a spinning phase defect in the equatorial plane receives an additional contribution:

$$\delta\alpha_{Kerr} = -(4GJ)/(c^3 b^2) \sin(\varphi), \quad (45.45)$$

where  $\varphi$  is the azimuthal angle of the ray relative to the spin axis. This matches the GR prediction for lensing by a Kerr spacetime metric (Sereno 2003) at leading order, derived here purely from the off-diagonal admissibility tensor without invoking the Kerr solution of the Einstein equations.

---

## 29. Wave Optics and Coherence in Phase Lensing

The geometric optics approximation underlying the phase-cost variational principle of Proposition 45.2 is valid in the limit where the wavelength  $\lambda$  of the phase mode is much smaller than the Fresnel scale  $r_F = \sqrt{(\lambda D_{\text{eff}})}$ , where  $D_{\text{eff}} = D_L D_S / D_{LS}$  is the effective lens distance. When this condition is violated—for example, for radio waves lensed by stellar-mass objects, or for gravitational waves (with frequencies 10–1000

Hz) lensed by galaxies—the wave nature of the phase modes becomes essential and the ray picture breaks down.

In Phase Theory, wave optics is more fundamental than geometric optics: the phase modes are intrinsically wave-like (they are phase patterns on the substrate, not point particles), and geometric optics is the short-wavelength approximation to the wave propagation equation. The full wave-optics treatment of lensing uses the Kirchhoff-Fresnel diffraction integral to compute the amplitude at the observer:

$$A(\theta_{obs}) \propto \int \exp(i \omega \tau(\theta)) d^2\theta, \quad (45.46)$$

where  $\omega$  is the mode frequency and  $\tau(\theta)$  is the phase arrival time surface (45.26). The intensity is  $|A|^2$ , and images correspond to stationary points of the integrand (saddle points of  $\tau$  at frequency-independent positions, confirming the geometric optics limit  $\omega \rightarrow \infty$ ).

The Coherence Limitation of Axiom 4 modifies the Kirchhoff integral (45.46) near caustics. At the level of the path integral, the mutual phase information between paths separated by more than the coherence length  $l_{coh}$  in the lens plane is suppressed by the exponential factor  $\exp(-|\theta_1 - \theta_2|^2/l_{coh}^2)$ . This effectively cuts off the integral at a maximum transverse scale  $l_{coh}$ , modifying the interference pattern near caustics. The resulting magnification profile near a fold caustic in Phase Theory is:

$$\mu_{fold}(\Delta\theta) = |A_{fold}(\Delta\theta)|^2 \times \exp(-2\Delta\theta^2/l_{coh}^2\theta_E^2), \quad (45.47)$$

where  $A_{fold}(\Delta\theta)$  is the standard Airy function result for fold caustic magnification and the exponential is the coherence suppression factor. For  $l_{coh} \gg \theta_E$ , the suppression is negligible and the Phase Theory prediction equals the GR wave-optics result. For  $l_{coh} \sim \theta_E$ , observable suppression of caustic magnification peaks arises—a unique prediction of Phase Theory (Prediction 45.3, Section 35).

---

## 30. Gravitational Wave Lensing as Phase Lensing

Gravitational waves in Phase Theory are coherent oscillations of the phase stiffness tensor  $K_{ab}(x)$ —tensor phase modes that represent collective distortions of the

substrate's stiffness rather than of the phase field  $\Phi$  itself. Their propagation is governed by the same phase-cost variational principle as massless scalar phase modes (light), because both derive from the same Lagrangian density of the phase-inconsistency functional  $I[\Phi]$ . The gravitational wave modes are spin-2 (they deform the phase stiffness tensor with quadrupole symmetry), but their propagation equation at the geometric optics level is identical to equation (45.9).

The equality of the gravitational wave propagation speed with the speed of light,  $v_{\text{GW}} = c$ , follows in Phase Theory from the fact that both electromagnetic and gravitational phase modes travel at the maximum admissible reconfiguration speed  $c = \sqrt{(K_0/\rho_0)}$ . Both are massless in the topological sense (neither carries net topological charge), and both propagate through the same admissibility landscape  $\eta(\mathbf{x})$ . The GW170817 measurement constraining  $|v_{\text{GW}} - c|/c < 10^{-15}$  is thus automatically satisfied in Phase Theory: there is no mechanism by which gravitational and electromagnetic modes could travel at different speeds in the vacuum, since they are governed by the same substrate parameters.

Gravitational waves are gravitationally lensed by exactly the same admissibility gradients as light. The deflection angle for a gravitational wave is identical to that for a light ray of the same frequency at the same impact parameter:  $\alpha_{\text{GW}} = \alpha_{\text{EM}} = 4GM/(c^2b)$ . This is the Phase Theory realization of the equivalence principle for phase modes. In GR, this equality follows from the coupling of all forms of energy to the same spacetime metric; in Phase Theory, it follows from the fact that all massless phase modes (regardless of spin) propagate through the same admissibility landscape.

When a gravitational wave signal is strongly lensed by a massive object, multiple GW images arrive at the detector with time delays given by equation (45.35). Since the GW signal carries phase information from the source (the binary inspiral waveform), the two lensed images are coherent copies of the same signal, delayed by  $\Delta t$ . If the frequency of the GW signal  $\omega$  is such that  $\omega\Delta t \sim 2\pi$ , the two images interfere constructively or destructively, producing frequency-dependent magnification (beating patterns). The beating frequency is  $\Delta f_{\text{beat}} = 1/\Delta t$ , and the amplitude of the beating pattern encodes both the time delay and the flux ratio of the images.

Current searches for lensed GW events in the LIGO/Virgo O3 dataset (Abbott et al. 2023, LVK lensing search) have found no confirmed strongly lensed GW events, consistent with the expected rate of  $\sim 1\text{--}3\%$  of detected events being strongly lensed at O3 sensitivity. Phase Theory predicts that the first confirmed strongly lensed GW events will be detected during O4/O5 operations (expected 2024–2026), with wave-optics beating patterns that probe the lens phase-density profile at  $\sim 10^8 M_\odot$  mass scales.

---

### 31. Plasma Lensing and Dispersive Effects

In astrophysical contexts, massless phase modes propagating along a line of sight pass not only through the admissibility gradients of massive defects but also through regions containing free electrons (ionized plasma). In GR, plasma introduces a frequency-dependent refractive index  $n_{\text{plasma}}(\nu) = \sqrt{(1 - \nu_p^2/\nu^2)} \approx 1 - \nu_p^2/(2\nu^2)$ , where  $\nu_p = \sqrt{(n_e e^2/(\pi m_e))}$  is the plasma frequency and  $n_e$  is the electron number density. This causes chromatic (frequency-dependent) deflection of radio waves, known as plasma lensing.

In Phase Theory, plasma is a distribution of charged phase defects (electrons, classified by  $\Xi = (k=0, Q_H=1/2, t, \chi=1/2, R=1)$ ), and their collective effect on propagating phase modes is an additional admissibility contribution  $\eta_e(x, \nu)$  that depends on both position and frequency. The total admissibility index experienced by a phase mode of frequency  $\nu$  propagating through a plasma-filled gravitational lens is:

$$\eta_{\text{total}}(x, \nu) = \eta_{\text{grav}}(x) + \eta_{\text{plasma}}(x, \nu) = [1 + \Phi_N/c^2] + [-\nu_p^2(x)/(2\nu^2)]. \quad (45.48)$$

The Phase Theory prediction differs from linear superposition of gravitational and plasma effects at second order. Because both contributions to  $\eta_{\text{total}}$  enter the same phase-cost functional, their cross-term in the second-order expansion produces an interaction:

$$\delta\eta_{\text{cross}}(x, \nu) \sim \Phi_N \times \nu_p^2/\nu^2 = -(GM/c^2 r) \times (\nu_p^2/\nu^2), \quad (45.49)$$

which is absent in the GR treatment, where gravitational and plasma lensing are computed independently and added. For typical pulsar timing array targets near plasma-rich lens systems (e.g., pulsars in the Galactic center or near the Galactic plane), this cross-term produces a frequency-dependent correction to the dispersive measure (DM) of order  $\delta\text{DM}/\text{DM} \sim (r_s/r) \sim 10^{-6}\text{--}10^{-5}$ —potentially detectable with the Square Kilometre Array (SKA) timing precision of  $\sim 10$  ns over multi-year baselines.

---

## 32. Self-Lensing and Phase Feedback

A massive phase defect (compact object) can gravitationally lens its own emitted phase modes, creating a situation in which the admissibility gradient of the defect acts on the modes that it itself produces. This is the Phase Theory description of the photon sphere: the radius at which massless phase modes can orbit the defect in an unstable circular orbit, continuously bending their propagation path back toward the defect.

The condition for a circular phase-mode orbit at radius  $r$  is that the centripetal admissibility gradient exactly equals the requirement for circular motion at that radius. From the phase-cost ray equation (45.9), this condition is:

$$d/dr [r / \eta(r)] = 0, \quad (45.50)$$

which for  $\eta(r) = (1 - r_s/r)^{-1/2}$  (the strong-field profile, not the weak-field approximation) gives  $r_{\text{ph}} = (3/2)r_s$ . This is the same photon sphere radius as in GR, here derived from the condition on the admissibility profile without invoking the Schwarzschild solution of the Einstein equations.

The photon sphere orbit is unstable: a phase mode displaced slightly inward will spiral into the defect core, and one displaced slightly outward will escape. This instability is a property of the admissibility profile, specifically the fact that  $d^2/dr^2[r/\eta(r)] > 0$  at  $r = r_{\text{ph}}$ . In Phase Theory, the coherence limitation (Axiom 4) modifies the photon sphere condition at scales below the coherence length  $l_{\text{coh}}$ . At the radius  $r_{\text{ph}}$ , the phase-mode orbital "length"  $2\pi r_{\text{ph}}$  must be much larger than the coherence length  $l_{\text{coh}}$  for the orbit condition to be self-consistent. For macroscopic

black holes,  $2\pi r_{\text{ph}} \sim 10 \text{ km}$  (for a solar-mass black hole)  $\gg l_{\text{coh}} \sim l_{\text{Planck}} \sim 10^{-35} \text{ m}$ , so the correction is negligible. For near-Planck-scale defects, the correction shifts  $r_{\text{ph}}$  and modifies the shadow angular size—a potential signature in Planck-scale compact object lensing (should such objects exist at detectable abundance).

The photon sphere defines the boundary between two qualitative regimes of lensing by compact phase defects. Phase modes with impact parameter  $b > r_{\text{ph}}\sqrt{3}/r_s$  (specifically  $b > b_{\text{crit}} = (3\sqrt{3}/2) r_s$ ) escape to infinity after deflection. Phase modes with  $b < b_{\text{crit}}$  are captured by the defect (in GR, they fall into the black hole; in Phase Theory, they are absorbed into the defect core). The boundary  $b = b_{\text{crit}}$  corresponds to the photon sphere orbit, and the bright ring of emission observed in the EHT image of M87\* traces the pile-up of escaping photon trajectories near this critical impact parameter.

---

### 33. Lensing Selection Effects and Survey Biases

Gravitational lensing affects not only the shapes and positions of background sources but also their apparent fluxes and the probability that they are included in a flux-limited survey. These selection effects must be carefully accounted for in the interpretation of any survey that uses gravitationally lensed sources.

Magnification bias arises because lensing preferentially brings sources that are intrinsically too faint to be included in a flux-limited survey above the detection threshold. The effect depends on the slope of the source number counts at the threshold flux: for a steep number count slope  $n(>F) \sim F^{-\beta}$  with  $\beta > 1$ , lensing increases the observed surface density of sources (the magnified sources more than compensate for the dilution of the sky area); for  $\beta < 1$ , lensing decreases the surface density. This magnification bias is identical in Phase Theory and GR, since it depends only on the flux magnification factor  $\mu = 1/\det(A)$ , which is the same in both theories at leading order.

Submillimeter galaxies (SMGs) at redshifts  $z \sim 2\text{--}4$  are particularly subject to magnification bias, because they have steep number counts and exist in an era of peak star formation. The brightest SMGs detected by surveys such as

Herschel/SPIRE and SCUBA-2 are typically gravitationally lensed by a factor  $\mu \sim 5\text{--}50$ , bringing them above the sensitivity threshold of these surveys. Phase Theory predicts the same lensed SMG population as GR at leading order, with corrections at the sub-percent level from higher-order admissibility effects.

Fast radio bursts (FRBs) are millisecond-duration radio transients of extragalactic origin (Lorimer et al. 2007; Thornton et al. 2013) that provide a novel probe of lensing on cosmological scales. The short duration ( $\Delta t \sim \text{ms}$ ) makes FRBs sensitive to wave-optics lensing by compact objects with Einstein ring radii smaller than the Fresnel scale for radio wavelengths. Phase Theory predicts a specific modification of the FRB scintillation bandwidth  $\Delta\nu_{\text{scint}}$  near lens systems: the coherence limitation (Axiom 4) reduces the scintillation bandwidth by a factor  $(1 - r_F^2/l_{\text{coh}}^2)$  relative to the geometric-optics prediction, potentially observable with next-generation radio arrays (DSA-2000, CHORD, SKA) that can detect and localize FRBs at rates of thousands per day.

---

## 34. Precision Tests of the Lensing Deflection Law

The classical Eddington measurement of 1919 established the solar light deflection at the  $\sim 30\%$  precision level. Subsequent decades of VLBI measurements of the apparent positions of extragalactic radio sources near the Sun have refined this measurement by orders of magnitude. The current precision record is held by the VLBI measurement of Shapiro et al. (2004), which achieved  $\sim 0.02\%$  accuracy in the solar deflection:  $\alpha_{\text{obs}}/\alpha_{\text{GR}} = 0.99983 \pm 0.00045$ . This is fully consistent with both GR and Phase Theory, since both predict identical deflection angles at this level of precision.

The parameterized post-Newtonian (PPN) framework encodes the solar system predictions of generic metric theories of gravity in terms of parameters  $\gamma, \beta, \xi, \dots$ . The parameter most directly constrained by lensing is  $\gamma$ , defined by the isotropic weak-field metric:  $g_{00} = 1 - 2GM/c^2r$ ,  $g_{ij} = (1 + 2\gamma GM/c^2r) \delta_{ij}$ . GR predicts  $\gamma = 1$ . Phase Theory at leading order also predicts  $\gamma = 1$ , since the phase stiffness tensor is isotropic in the weak-field limit and the spatial admissibility gradient equals the

temporal gradient exactly (Proposition 45.3). Deviations from  $\gamma = 1$  in Phase Theory arise only at order  $(l_{\text{coh}}/r_s) \times (r_s/r) \sim 10^{-37}$  at solar system scales—immeasurably small.

The Cassini measurement (Bertotti et al. 2003) achieved  $|\gamma - 1| < 2.3 \times 10^{-5}$  from multi-frequency Doppler tracking of the Cassini spacecraft during solar superior conjunction. This constrains Phase Theory's coherence correction parameter to  $\epsilon_{\text{coh}} < 2.3 \times 10^{-5} \times (b/r_s) \sim 10^{-7}$  for solar system impact parameters  $b \sim R_\odot$ , a constraint that is readily satisfied and places no meaningful bound on  $l_{\text{coh}}$ .

Gaia astrometric lensing represents the next frontier. The Gaia mission (Gaia Collaboration 2016, 2018, 2023) measures stellar positions to sub-microarcsecond precision for over  $10^9$  stars. At this precision, the apparent positions of background stars are shifted by lensing due to solar system objects (Jupiter: up to 1.6 mas, Saturn: up to 0.58 mas) and nearby high-proper-motion stars (Sahu et al. 2017; Zurlo et al. 2018). Phase Theory predicts deflection angles for all solar system bodies and nearby stars that are identical to GR at Gaia's precision level, since the Phase Theory corrections are of order  $(r_s/b)^2 \sim 10^{-15}$ – $10^{-12}$ , far below Gaia's  $10^{-6}$  arcsecond precision.

---

## 35. Predictions Unique to Phase Theory Lensing

The formal equivalence between Phase Theory and GR lensing at first post-Newtonian order (proved in Section 37) means that discriminating tests require measurements either at second or higher post-Newtonian order, at scales comparable to the coherence length  $l_{\text{coh}}$ , or in regimes where the topological structure of Phase Theory's dark sector (neutral topological sectors) differs observably from the GR dark matter description. We present here the four primary unique predictions of Phase Theory for gravitational lensing, with quantitative estimates of their magnitudes and observational feasibility.

**Prediction 45.3 (Coherence Suppression of Lensing Near Caustics).**



At impact parameters within the coherence length  $l$

coh

of a caustic, magnification is suppressed relative to the geometric optics prediction by:

$$\mu_{PT} = \mu_{GR, wave-optics} \times \exp(-\Delta\tau/\tau_{coh}), \quad (45.51)$$

where  $\Delta\tau$  is the phase arrival time difference between the interfering image paths and  $\tau$

coh

=  $l$

coh

$/c$  is the coherence time. This differs from the GR wave-optics prediction (which uses the standard Kirchhoff integral without an exponential coherence cutoff) by the factor  $\exp(-\Delta\tau/\tau$

coh

).

#### **Prediction 45.4 (Neutral-Sector Lensing Power Spectrum Cutoff).**

The weak lensing convergence power spectrum has a characteristic suppression at multipoles:

$$C_k(\ell)_{PT} = C_k(\ell)_{GR} \times [1 - \exp(-\ell_{min}^2/\ell^2)], \quad \ell \gg \ell_{min} = D_L / L_{coh}, \quad (45.52)$$

arising from the minimum defect size  $M$

min

of the neutral topological sector. This produces a cutoff in the weak lensing power

spectrum at  $\ell \sim \ell$

min

= D

L

/l

coh

that differs from the warm dark matter (WDM) suppression, which has a different functional form (a power-law suppression rather than exponential).

#### **Prediction 45.5 (GW Lensing Waveform Asymmetry).**

Strongly lensed gravitational wave events show a waveform asymmetry between images due to the phase-cost temporal component. The asymmetry parameter is:

$$A_{\text{asym}} = (\Delta t_A - \Delta t_B)_{PT} / (\Delta t_A - \Delta t_B)_{GR} - 1 = O(r_s/b) \sim 10^{-4}. \quad (45.53)$$

This asymmetry arises from the different temporal ordering cost experienced by the two images due to their different impact parameters, and is absent at the level of accuracy of the GR time delay formula.

#### **Prediction 45.6 (Refractive Deviation in Photon Ring).**

The photon ring angular diameter for a black hole of mass M at distance D is:

$$\theta_{\text{ring}} = (1 + \varepsilon_{\text{coh}}) (3\sqrt{3} GM) / (c^2 D), \quad \varepsilon_{\text{coh}} = O(l_{\text{coh}}/r_s). \quad (45.54)$$

For M87\* ( $M = 6.5 \times 10$

9

M

⊙

,  $D = 16.8 \text{ Mpc}$ ), the correction  $\varepsilon$

coh

$\times \theta$

ring

$\sim 10 \text{ nanoarcseconds}$ , at the threshold of ngEHT sensitivity ( $\sigma$

$\theta$

$\sim 1 \text{ } \mu\text{as}$  for the 2030 ngEHT configuration).

We emphasize that Predictions 45.3–45.6 are not free parameters that can be tuned: they are consequences of Phase Theory's fundamental axioms (specifically Axiom 4 for 45.3 and 45.6; Axiom 3 for 45.4; Axiom 2 for 45.5), and their magnitudes are determined by the single coherence scale  $l_{\text{coh}}$ . If any one of these predictions is confirmed, it constrains  $l_{\text{coh}}$ ; if any one is falsified (i.e., the Phase Theory prediction is observed to be inconsistent with observations at the required precision), it places a bound on  $l_{\text{coh}}$  that may be tested against the other predictions. The coherence length  $l_{\text{coh}}$  is thus the primary free parameter of Phase Theory's lensing phenomenology, playing a role analogous to the Compton wavelength in quantum gravity phenomenology.

---

## 36. Experimental Protocols for Phase Lensing Tests

**Protocol 1: Next-Generation Event Horizon Telescope (ngEHT).** The ngEHT collaboration aims to achieve an angular resolution of  $\sim 1 \text{ } \mu\text{as}$  at 345 GHz by 2030, through the addition of new stations in Africa, Asia, and the Pacific, and through

enhanced receiver technology and bandwidth. At this resolution, the photon ring angular diameter of M87\* ( $\sim 42 \mu\text{as}$ ) can be measured to  $\sim 0.1 \mu\text{as}$  precision (0.3%), enabling a direct test of Prediction 45.6. Phase Theory predicts a deviation of  $\sim 10$  nanoarcseconds from the GR photon ring diameter, which lies at the threshold of ngEHT sensitivity. The primary challenge is separating the coherence correction from systematic errors in the source model (non-uniform accretion flow emission) and station calibration. A clean test requires multiple epochs and multi-frequency observations to characterize the accretion structure independently.

**Protocol 2: LIGO/Virgo O4-O5 Gravitational Wave Lensing Search.** The O4 run (beginning 2023) and O5 run (projected 2025–2027) will detect GW events at rates of  $\sim 200$  and  $\sim 600$  per year respectively. The expected fraction of strongly lensed events is  $\sim 0.1$ –1% at O4 sensitivity and  $\sim 1$ –5% at O5 sensitivity, corresponding to  $\sim 1$ –30 lensed events per run. Detection of the beating pattern in the GW amplitude as a function of frequency tests the wave-optics nature of the lensing (Section 30) and measurement of the waveform asymmetry (at  $O(10^{-4})$  level) tests Prediction 45.5. The latter requires signal-to-noise ratios  $\gtrsim 10^4$  in individual GW events, achievable at O5 sensitivity for nearby events.

**Protocol 3: Euclid/LSST Weak Lensing Power Spectrum.** The Euclid satellite (launched 2023) and the Vera C. Rubin Observatory LSST (first light 2024) will together measure the weak lensing shear power spectrum to multipoles  $\ell \sim 3 \times 10^3$ – $10^4$  with sub-percent precision. Phase Theory predicts a suppression of the lensing power spectrum at  $\ell > \ell_{\text{min}} = D_L/l_{\text{coh}}$  (Prediction 45.4). Detecting this suppression requires measuring the power spectrum to  $\ell \sim 10^4$ – $10^5$ , which is within the Stage IV survey design range. The primary contamination is intrinsic galaxy alignments and baryonic feedback effects (AGN-driven outflows, supernova feedback), which must be modeled and marginalized. A robust detection of the Phase Theory cutoff will require cross-validation with hydrodynamical simulations to isolate the dark matter small-scale physics.

**Protocol 4: SKA/DSA-2000 FRB Lensing Survey.** The Square Kilometre Array (SKA-mid, South Africa) and the Deep Synoptic Array (DSA-2000, USA) will detect FRBs at rates of  $\sim 10^4$ – $10^5$  per year with millisecond-precision localizations. A fraction  $\sim 0.01$ –0.1% of these are expected to be gravitationally lensed by

intervening mass concentrations, producing echo pulses delayed by  $\sim 1\text{--}1000$  ms. For FRBs near caustics of foreground lenses, the coherence suppression of Prediction 45.3 modifies the scintillation bandwidth by a factor  $\exp(-\Delta\tau/\tau_{\text{coh}})$ . Measurement of this suppression factor for a large sample of lensed FRBs at different source-caustic offsets can directly constrain  $l_{\text{coh}}$ .

**Protocol 5: VLBI Solar Deflection at Centimeter Wavelengths.** Continued VLBI monitoring of quasar positions near the Sun during solar conjunctions can improve the precision on  $\gamma$  to  $\sim 10^{-7}$  with the next generation of radio arrays (ngVLA, SKA-VLBI). At this precision, the Phase Theory second-order correction to the Shapiro delay (Lemma 45.1), of order  $(r_s/b)^2 \sim 10^{-12}$  for solar impact parameters, is still immeasurably small. However, the constraint on  $\gamma$  at  $10^{-7}$  would provide a strong bound on any scalar field modification of Phase Theory that could arise from higher-order terms in the admissibility expansion.

**Protocol 6: CMB-S4 Lensing Cross-Correlation.** The CMB-S4 experiment (projected 2027–2031) will produce a cosmic-variance-limited CMB lensing convergence map with angular resolution  $\sim 1'$ . Cross-correlating this map with the galaxy distribution from Euclid and LSST provides a measurement of the galaxy bias and the galaxy-lensing cross-correlation as a function of angular scale. Phase Theory predicts a scale-dependent galaxy bias enhancement near the coherence length scale that is absent in CDM: neutral-sector defects below the coherence length cannot cluster at those scales, modifying the matter-bias relationship. Detection of this scale dependence would simultaneously test Prediction 45.4 and the neutral-sector dark matter model.

**Protocol 7: Roman Space Telescope Strong Lensing Survey.** The Nancy Grace Roman Space Telescope (projected launch 2026) will conduct a deep infrared survey identifying  $\sim 10^5$  strong lensing systems (galaxy-galaxy lenses and group/cluster lenses). Time delays measured for a large subset of these systems (those with variable sources: quasars, supernovae) provide a direct test of Lemma 45.1, which predicts a positive offset in the Phase Theory time delays relative to GR at second order in  $r_s/b$ . A sample of  $\sim 10^3$  precisely measured time delays spanning a range of lens compactnesses  $r_s/b$  can determine the Phase Theory correction  $\varepsilon_{\text{coh}}$  with a

precision of  $\sim 1\%$  of the predicted correction, sufficient to discriminate from zero at  $3\sigma$  if  $\epsilon_{\text{coh}} = O(1)$ .

### 37. Comparison with General Relativity: Formal Equivalence and Deviations

| Observable                       | GR Explanation                                    | Phase Theory Explanation                         | Leading-Order Agreement                                   | First Deviation                                | Status                                                  |
|----------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------|------------------------------------------------|---------------------------------------------------------|
| Light deflection angle $\alpha$  | Null geodesic in Schwarzschild spacetime          | Phase-cost ray equation with $\eta(r)$           | Exact: $4GM/c^2b$                                         | $O((r_s/b)^2) \times \epsilon_{\text{coh}}$    | Confirmed to 0.02%                                      |
| Shapiro time delay               | Time-time metric component $g_{00}$               | Reduced phase-mode speed $c/\eta(x)$             | Exact: $(2GM/c^3)\ln(4r_{\text{ER}}/b^2)$                 | $O((r_s/b)^2) \times \epsilon_{\text{coh}}$    | Confirmed to 0.002% (Cassini)                           |
| Convergence map $\kappa(\theta)$ | Projected matter density / $\Sigma_{\text{crit}}$ | Projected phase density / $\Sigma_{\text{crit}}$ | Identical at first order                                  | Coherence cutoff at $\ell > \ell_{\text{min}}$ | Consistent to $\sim 5\%$ (Stage III surveys)            |
| Shear field $\gamma(\theta)$     | Tidal gravitational field                         | Tidal admissibility gradient                     | Identical at first order                                  | $O((r_s/r)^2)$ anisotropy correction           | Consistent to $\sim 3\%$ (KiDS, HSC, DES)               |
| Magnification $\mu$              | Liouville's theorem for photon geodesics          | Conservation of admissible phase paths           | Identical: $1/((1-\kappa)^2-\gamma^2)$                    | Coherence suppression near caustics            | Consistent (no caustic-crossing events at high S/N yet) |
| Einstein radius $\theta_E$       | Null geodesic crossing: $\alpha=\theta$ condition | Phase-cost deflection: $\alpha=\theta$ condition | Identical: $\sqrt{(4GMD_{\text{LS}}/c^2D_{\text{LD}_S})}$ | $O((r_s/b)^2)$ correction to ring radius       | Confirmed in $\sim 10^4$ systems (SLACS, BOSS, etc.)    |
| Time delay                       | Refsdal                                           | Phase-cost                                       | Identical at                                              | +0.3%                                          | Under                                                   |

| Observable                | GR Explanation                                 | Phase Theory Explanation                                                    | Leading-Order Agreement             | First Deviation                                                           | Status                                            |
|---------------------------|------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------|
| cosmography $H_0$         | formula, geodesic time delay                   | arrival time surface (45.26)                                                | first order                         | systematic from Lemma 45.1                                                | investigation ( $H_0$ tension)                    |
| Photon ring (EHT)         | Photon sphere in Kerr spacetime                | Phase-mode circular orbit in $\eta(r)$                                      | Identical: $r_{\text{ph}} = 3r_s/2$ | $O(l_{\text{coh}}/r_s)$ shift in ring diameter ( $\sim 10$ nas for M87*)  | Consistent with EHT 2019 (precision $\sim 10\%$ ) |
| GW lensing                | GW null geodesics same as photon geodesics     | Tensor phase modes follow same phase-cost equation                          | Identical deflection, equal speeds  | $O(r_s/b)$ waveform asymmetry                                             | No confirmed lensed GW yet (consistent)           |
| CMB lensing               | Null geodesics from $z=1100$                   | Phase modes from last scattering phase-cost path                            | Identical at first order            | $f_{\text{NL}} \sim O(1)$ modification to lensing bispectrum              | Consistent with Planck 2018                       |
| Caustic structure         | Fold/cusp singularities of geodesic focusing   | Fold/cusp singularities of phase-cost surface                               | Identical topology                  | Coherence suppression of peak magnification (Pred. 45.3)                  | Not yet tested at required precision              |
| Wave optics (GW/radio)    | Kirchhoff-Fresnel integral over null geodesics | Kirchhoff-Fresnel integral with coherence cutoff                            | Identical at low frequency          | $\exp(-\Delta\tau/\tau_{\text{coh}})$ suppression near caustics           | Not yet tested                                    |
| Dispersive/plasma effects | GR + plasma refractive index (separate)        | $\eta_{\text{total}} = \eta_{\text{grav}} + \eta_{\text{plasma}}$ (unified) | Leading terms identical             | Cross-term $\delta\eta_{\text{cross}} \sim O(r_s/r \times \nu_p^2/\nu^2)$ | Below current precision                           |
| Dark matter lensing       | CDM halos with NFW profile                     | Neutral topological sector halos                                            | NFW-like at intermediate scales     | Small-scale cutoff at $M_{\text{min}}$ (Pred. 45.4)                       | Consistent; cutoff not yet probed                 |

Table 2. Full comparison of GR and Phase Theory lensing predictions for fourteen observables.  
*"First Deviation" indicates where the two theories differ at lowest non-trivial order.*

### **Theorem 45.3 (First-Order Lensing Equivalence).**

At first post-Newtonian order  $O(r$

$s$

$/b)$ , all lensing observables predicted by Phase Theory are identical to those predicted by general relativity with the same matter distribution.

Proof sketch.

We establish the equivalence by showing that Phase Theory's phase-cost variational principle (45.6) reduces to the null geodesic equation of GR in isotropic coordinates at first post-Newtonian order.

Step 1: Write the isotropic Schwarzschild metric in the weak-field limit:  $ds^2 = -(1 - r_s/r)c^2 dt^2 + (1 + r_s/r)(dx^2 + dy^2 + dz^2) + O((r_s/r)^2)$ . The null condition  $ds^2 = 0$  gives:  $(1 + r_s/r)|d\mathbf{x}/dt|^2 = (1 - r_s/r)c^2$ , so the spatial ray speed is  $v = c(1 - r_s/r)(1 + r_s/r)^{-1/2} \approx c(1 - 2r_s/(2r)) = c/\eta(r)$  where  $\eta(r) = 1 + r_s/(2r) + O((r_s/r)^2) = 1 + \Phi_N/c^2 + O((r_s/r)^2)$ . This is precisely the Phase Theory admissibility index of Proposition 45.1.

Step 2: The null geodesic equation in isotropic coordinates gives a deflection angle identical to the phase-cost ray equation (45.9) at first order. Both yield  $\alpha = 4GM/(c^2 b)$ . Step 3: The Shapiro delay follows identically from  $v(x) = c/\eta(x)$  in both frameworks. Step 4: All derived quantities (convergence, shear, magnification, Einstein radius, time delay) are computed from  $\alpha(\theta)$  and  $\eta(x)$ , which are identical at first order. Therefore all lensing observables agree at first post-Newtonian order.  $\square$

Beyond first order, the two theories differ. The deviations are characterized by the dimensionless parameter  $\varepsilon = (r_s/b) \times (l_{\text{coh}}/r_s)$  for the coherence corrections, and by



$(r_s/b)^2$  for the nonlinear admissibility corrections. Both classes of correction are small in current observational contexts but potentially measurable with next-generation experiments (Section 36).

---

## 38. Implications for the Ontology of Gravity

The central result of this paper is that gravitational lensing—the canonical exhibit of spacetime curvature—requires no spacetime curvature for its derivation. The full quantitative agreement between Phase Theory and GR at first post-Newtonian order, established in Theorem 45.3, demonstrates that the curved-spacetime description is not the only possible physical account of lensing; it is an effective description, valid on scales large compared to  $l_{\text{coh}}$ , of a more fundamental refractive mechanism operating at the level of the phase substrate.

This has profound ontological implications. The standard narrative of the 1919 Eddington experiment is that it proved that space itself is curved. Phase Theory shows that this conclusion is unwarranted by the data: what the data prove is that light deflects by  $4GM/(c^2b)$  near the Sun, not that the cause of this deflection is the curvature of a spacetime manifold. Any theory that produces a spatially varying propagation speed for massless modes, with both spatial and temporal components equal to  $GM/(c^2b)$ , will reproduce this observation. Phase Theory is such a theory, and it does not invoke a spacetime manifold at all.

The equivalence principle receives a natural reinterpretation in Phase Theory. In GR, the equivalence principle is the statement that all forms of mass-energy couple to the same spacetime metric, which causes all objects to follow the same geodesics regardless of their internal composition. The universality of gravity is therefore a consequence of the universality of the spacetime manifold. In Phase Theory, there is no spacetime manifold, and the equivalence principle is instead a consequence of the fact that all massless phase modes (regardless of their internal structure: electromagnetic, gravitational, or other) propagate through the same admissibility landscape  $\eta(x)$ . Since  $\eta(x)$  is a property of the substrate, not of the modes, it affects

all modes identically. Gravity is universal not because of geometry, but because the substrate is the common medium through which all physics occurs.

The question "does gravity bend space?" dissolves completely in Phase Theory. There is no space to bend: what we call "space" is the emergent macroscopic structure of the phase substrate, described by the metric tensor  $g_{\mu\nu} = \alpha \langle \nabla \Phi \nabla \Phi \rangle_{\text{coh}}$  of Axiom 5. Asking whether gravity bends space is like asking whether heating bends the temperature field: the "bending" of the temperature field is an epiphenomenon of the underlying physics (molecular motion), not a primitive property of a geometric object. Analogously, the "bending" of spacetime by gravity is an epiphenomenon of the underlying phase-substrate dynamics, not a property of a geometric manifold that exists prior to and independently of the matter and energy content of the universe.

The refractive picture of gravity offered by Phase Theory is not merely conceptually more transparent—it is empirically richer. It predicts a set of observable consequences (Predictions 45.3–45.6) that are absent from GR, and it provides a unified account of dark matter (as neutral topological sectors generating admissibility gradients) that eliminates the need for dark matter as a separate species of matter to be introduced *ad hoc* into the theory. The program for the next decade of gravitational lensing science, outlined in the experimental protocols of Section 36, is simultaneously a program for testing the foundational axioms of Phase Theory. Each protocol addresses a specific axiom: Protocol 1 tests the coherence saturation prediction of Axiom 4; Protocol 2 tests the equal-speed prediction of Axioms 1 and 2; Protocol 3 tests the topological stability of neutral sectors (Axiom 3); Protocol 4 tests the coherence limitation of Axiom 4; Protocols 5–7 test the metric-emergence prediction of Axiom 5.

In summary: gravitational lensing is not the bending of light by curved spacetime. It is the refraction of massless phase modes through a spatially varying admissibility landscape, induced by the presence of stable topological defects in the pre-geometric phase substrate. This is a more fundamental, more physical, and more explanatory account of one of the most celebrated phenomena in modern astrophysics. The curved spacetime description of GR remains an excellent effective theory at first post-Newtonian order, but it is not the correct ontological description

—and the next generation of gravitational lensing experiments will determine whether Phase Theory's deeper description is the correct one.

---

## 39. Appendix A — Phase-Cost Geodesic Equations

### 39.1 Full Derivation of the Euler-Lagrange Equations

We present the complete derivation of the phase-cost Euler-Lagrange equations from the functional  $C[\gamma] = \int_0^L \eta(\gamma(s)) ds$ , where  $\gamma(s)$  is a curve parametrized by arc length. The variation  $\gamma(s) \rightarrow \gamma(s) + \varepsilon \zeta(s)$  with  $\zeta(0) = \zeta(L) = 0$  gives:

$$\delta C = \int_0^L [\nabla \eta \cdot \zeta + \eta d\zeta/ds \cdot dx/ds] ds = \int_0^L [\nabla \eta \cdot \zeta - d/ds(\eta dx/ds) \cdot \zeta] ds, \quad (45.55)$$

after integration by parts. Setting  $\delta C = 0$  for all  $\zeta$  gives the Euler-Lagrange equation:

$$d/ds[\eta(x) dx^\mu/ds] = \partial^\mu \eta(x). \quad (45.56)$$

Expanding the left side:  $\eta d^2x^\mu/ds^2 + (\partial_\nu \eta)(dx^\nu/ds)(dx^\mu/ds) = \partial^\mu \eta$ . Dividing by  $\eta$  and using the arc-length constraint  $dx^\mu/ds dx_\mu/ds = 1$ :

$$d^2x^\mu/ds^2 = (\partial^\mu_\nu - \dot{x}^\mu \dot{x}_\nu) \partial^\nu \ln \eta(x). \quad (45.57)$$

This confirms equation (45.9) of Section 8.

### 39.2 Comparison with Null Geodesic Equation in Isotropic Coordinates

The null geodesic equation in isotropic coordinates ( $x^\mu = (t, x^i)$ ) with metric  $ds^2 = -(1+2\Phi_N/c^2)c^2 dt^2 + (1-2\Phi_N/c^2)(dx^2+dy^2+dz^2)$  is:

$$d^2x^\mu/d\lambda^2 + \Gamma^\mu_{\nu\rho}(dx^\nu/d\lambda)(dx^\rho/d\lambda) = 0, \quad (45.58)$$

with the null constraint  $g_{\mu\nu}(dx^\mu/d\lambda)(dx^\nu/d\lambda) = 0$ . At first post-Newtonian order, the Christoffel symbols reduce to  $\Gamma^i_{00} = \partial_i \Phi_N/c^2$ ,  $\Gamma^i_{jk} = -\delta_{ij} \partial_k \Phi_N/c^2 + \dots$ , and the null geodesic equation for the spatial components becomes  $d^2x^i/dt^2 = -2\partial_i \Phi_N/c^2$ , which is exactly (45.57) with  $\nabla \ln \eta = \nabla(\Phi_N/c^2)$ , confirming Theorem 45.3.

### 39.3 Post-Newtonian Expansion to Second Order

The second-order post-Newtonian expansion of the phase-cost geodesic equations introduces terms of order  $(\Phi_N/c^2)^2$ . From the exact admissibility profile  $\eta(r) = (1 - r_s/r)^{-1/2}$ , the phase-cost ray equation at second order gives a deflection angle:

$$\alpha_{2PN} = (4GM/c^2 b)[1 + (15\pi/16)(r_s/b) + O((r_s/b)^2)]. \quad (45.59)$$

The second-order GR result (from the Schwarzschild metric geodesic) is  $\alpha_{GR,2PN} = (4GM/c^2 b)[1 + (15\pi/16)(r_s/b) + \dots]$ , identical to (45.59). The coherence correction enters at order  $\varepsilon_{coh}(r_s/b)$  beyond this, and constitutes the distinguishing Phase Theory prediction at second post-Newtonian order.

---

## 40. Appendix B — The Admissibility Transfer Equation

### 40.1 Radiative Transfer Analog for Phase Modes

In standard radiative transfer, the specific intensity  $I_\nu(\mathbf{x}, \mathbf{n})$  (energy per unit time, area, solid angle, and frequency) satisfies the Boltzmann equation along a ray:

$$dI_\nu/ds = -\kappa_\nu I_\nu + j_\nu, \quad (45.60)$$

where  $\kappa_\nu$  is the absorption coefficient and  $j_\nu$  is the emission coefficient. In the absence of absorption and emission (vacuum propagation through a lensing medium),  $dI_\nu/ds = 0$  along rays, expressing the conservation of surface brightness.

The Phase Theory analog replaces the specific intensity with the specific phase-mode occupation number  $f(\mathbf{x}, \mathbf{k})$  (number of phase modes per unit phase space volume), and the rays with phase-cost geodesics  $\gamma(s)$ . The phase Boltzmann equation in the absence of defect interactions is:

$$df/ds + \{H_\eta, f\} = 0, \quad (45.61)$$

where  $\{\cdot, \cdot\}$  is the Poisson bracket with respect to the phase-cost Hamiltonian  $H_\eta = \eta(\mathbf{x})|\mathbf{k}|$  (the phase-cost of a mode at position  $\mathbf{x}$  with wave vector  $\mathbf{k}$ ).

## 40.2 Conservation of Phase-Mode Occupation Number

Equation (45.61) implies that the phase-mode occupation number is conserved along phase-cost geodesics:  $f$  is a constant of motion of the Hamiltonian flow generated by  $H_\eta$ . This is the Phase Theory analog of Liouville's theorem: phase space volume is preserved by the Hamiltonian flow, and therefore  $f$  is conserved along trajectories. The physical content is that gravitational lensing (which deforms the ray trajectories) does not create or destroy phase modes; it only redirects them. The conservation of  $f$  along rays is equivalent to the conservation of surface brightness (Theorem 45.2), confirming the consistency of the Phase Theory lensing formalism.

## 40.3 Connection to Invariance of Surface Brightness

The surface brightness  $I_\nu$  is related to the phase-mode occupation number by  $I_\nu = (2h\nu^3/c^2) f$ , where  $h$  is Planck's constant. Since  $\nu$  is not invariant along lensed rays (it is shifted by the cosmological redshift), the conserved quantity is  $I_\nu/\nu^3 = (2h/c^2) f$ —the occupation number itself. This is the standard result that surface brightness is conserved in gravitational lensing (modulo cosmological redshift), expressed here as a consequence of the phase Liouville theorem (45.61). The detailed balance between magnification and beam compression is automatic in Phase Theory: Hamiltonian flow preserves phase space volume, and this preservation is what makes the lens mapping area-preserving (in the signed sense, i.e.,  $\det(A)$  encodes the signed area preservation).

---

## 41. Glossary of Phase Lensing Terms

**Admissibility gradient:** The spatial derivative  $\nabla\eta(x)$  of the admissibility index field. The admissibility gradient is the Phase Theory analog of the gravitational field, and it is the direct cause of phase refraction. See Section 6 and Theorem 45.1.

**Admissibility index:**  $\eta(x) = \sqrt{(K_0/K(x))}$ , the ratio of the vacuum phase stiffness to the local phase stiffness at substrate point  $x$ . Encodes the local cost of phase reconfiguration; analogous to the optical refractive index. See Definition 45.2 and Proposition 45.1.

**Coherence suppression:** The reduction in magnification near caustics due to the coherence limitation (Axiom 4), which bounds mutual phase information between substrate regions by their boundary area. Produces an exponential suppression factor  $\exp(-\Delta\tau/\tau_{\text{coh}})$  absent from the GR wave-optics calculation. See Prediction 45.3 and equation (45.51).

**Neutral topological sector:** A stable phase defect classified by  $\Xi = (k_{\text{em}}=0, Q_{\text{H}}=0, t, \chi, R_{\text{dark}})$  that is electrically neutral and electromagnetically invisible. Identified as the Phase Theory analog of dark matter. Produces admissibility gradients (gravitational lensing) without electromagnetic emission. See Section 21.

**Phase arrival surface:** The scalar field  $\tau(\theta) = (D_{\text{L}}D_{\text{S}})/(2cD_{\text{LS}})|\theta-\beta|^2 - \psi(\theta)$  over the image plane, whose stationary points determine image positions. The Phase Theory analog of the GR Fermat potential. See equation (45.26).

**Phase-cost functional:**  $C[\gamma] = \int_{\gamma} \eta(x) ds$ , the total admissibility cost of a trial path  $\gamma$  for a massless phase mode. Stationary paths of  $C[\gamma]$  determine the physical propagation paths of light and gravitational waves. See equation (45.7) and Proposition 45.2.

**Phase density:**  $\rho_{\text{phase}}(x) = \int W(x,y) |I_{\text{defect}}(y)| dV_y$ , the local concentration of admissible phase constraints induced by nearby stable defects. Not the same as mass density; rather, a property of the substrate at  $x$ . See Definition 45.1.

**Phase Einstein radius:** The angular radius  $\theta_{\text{E}}$  at which the admissibility deflection maps a trial image position to zero source offset. Numerically equal to the GR Einstein radius at leading order. See Definition 45.6.

**Phase lensing potential:**  $\psi(\theta)$ , the weighted line-of-sight integral of the admissibility excess  $(\eta-1)$ , whose gradient gives the reduced deflection angle. See Definition 45.3.

**Phase-mode propagation:** The propagation of massless collective phase modes (light, gravitational waves) through the substrate, governed by the phase-cost variational principle  $\delta \int \eta(x) ds = 0$ . Distinguished from massive defect propagation by

the absence of topological charge and the maximally fast reconfiguration speed. See Section 3.

**Phase refraction:** The continuous bending of phase-mode propagation directions in regions of spatially varying admissibility index  $\eta(x)$ , caused by the differential propagation speed  $v(x) = c/\eta(x)$  across the mode wavefront. The fundamental mechanism underlying gravitational lensing in Phase Theory. See Section 7.

---

## 42. References

1. Abbott, B.P. et al. (LIGO Scientific Collaboration and Virgo Collaboration). Observation of Gravitational Waves from a Binary Black Hole Merger. *Phys. Rev. Lett.* **116**, 061102 (2016).
2. Abbott, B.P. et al. (LIGO Scientific Collaboration and Virgo Collaboration and others). Multi-Messenger Observations of a Binary Neutron Star Merger. *Astrophys. J. Lett.* **848**, L12 (2017).
3. Abbott, R. et al. (LVK Collaboration). Search for Gravitational-Wave Signals Associated with Gamma-Ray Bursts during the Third Observing Run. *Astrophys. J.* **928**, 186 (2022).
4. Aubourg, É. et al. Cosmological implications of baryon acoustic oscillation measurements. *Phys. Rev. D* **92**, 123516 (2015).
5. Bartelmann, M. and Schneider, P. Weak gravitational lensing. *Phys. Rep.* **340**, 291 (2001).
6. Bekenstein, J.D. Relativistic gravitation theory for the modified Newtonian dynamics paradigm. *Phys. Rev. D* **70**, 083509 (2004).
7. Belokurov, V. et al. “The Cosmic Horseshoe”: Discovery of an Einstein Ring around a Giant Arc. *Astrophys. J. Lett.* **671**, L9 (2007).
8. Bertotti, B., Iess, L., and Tortora, P. A test of general relativity using radio links with the Cassini spacecraft. *Nature* **425**, 374 (2003).

9. Blandford, R.D. and Narayan, R. Fermat's principle, caustics, and the caustic crossing rates with applications to gravitational lensing. *Astrophys. J.* **310**, 568 (1986).
10. Clowe, D. et al. A Direct Empirical Proof of the Existence of Dark Matter. *Astrophys. J. Lett.* **648**, L109 (2006).
11. Collett, T.E. The Population of Galaxy-Galaxy Strong Lenses in Forthcoming Optical Imaging Surveys. *Astrophys. J.* **811**, 20 (2015).
12. Dalal, N. and Kochanek, C.S. Direct Detection of Cold Dark Matter Substructure. *Astrophys. J.* **572**, 25 (2002).
13. Diego, J.M. et al. JWST Observations of SMACS 0723: Strong Lensing Analysis and Mass Reconstruction. *Astron. Astrophys.* **665**, A134 (2022).
14. Dvali, G., Gabadadze, G., and Porrati, M. 4D Gravity on a Brane in 5D Minkowski Space. *Phys. Lett. B* **485**, 208 (2000).
15. Dyson, F.W., Eddington, A.S., and Davidson, C. A Determination of the Deflection of Light by the Sun's Gravitational Field, from Observations Made at the Total Eclipse of May 29, 1919. *Phil. Trans. R. Soc. A* **220**, 291 (1920).
16. Einstein, A. Zur allgemeinen Relativitätstheorie. *Sitzungsberichte der Preussischen Akademie der Wissenschaften* (1915).
17. Einstein, A. Die Grundlage der allgemeinen Relativitätstheorie. *Ann. Phys.* **49**, 769 (1916).
18. Event Horizon Telescope Collaboration. First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole. *Astrophys. J. Lett.* **875**, L1 (2019).
19. Event Horizon Telescope Collaboration. First Sagittarius A\* Event Horizon Telescope Results. I. The Shadow of the Galaxy's Central Black Hole. *Astrophys. J. Lett.* **930**, L12 (2022).
20. Gaia Collaboration. Gaia Data Release 3: Summary of the content and survey properties. *Astron. Astrophys.* **674**, A1 (2023).



21. Gorenstein, M.V., Shapiro, I.I., and Falco, E.E. Degeneracies in parameter extraction from gravitational lens data. *Astrophys. J.* **327**, 693 (1988).
22. Hanks, M. Phase Theory: Foundations of a Pre-Geometric Description of Physics (Paper 1). arXiv:2601.00001 [hep-th, gr-qc] (2026).
23. Hezaveh, Y.D. et al. Detection of Lensing Substructure Using ALMA Observations of the Dusty Galaxy SDP.81. *Astrophys. J.* **823**, 37 (2016).
24. Kaiser, N. and Squires, G. Mapping the dark matter with weak gravitational lensing. *Astrophys. J.* **404**, 441 (1993).
25. King, L.J. et al. The gravitationally lensed radio source B1938+666. *Mon. Not. R. Astron. Soc.* **295**, L41 (1998).
26. Kneib, J.-P. et al. Hubble Space Telescope observations of the lensing cluster Abell 2218. *Astrophys. J.* **471**, 643 (1996).
27. Lambert, S.B. and Le Poncin-Lafitte, C. Determining the relativistic parameter  $\gamma$  using Very Long Baseline Interferometry. *Astron. Astrophys.* **499**, 331 (2009).
28. Lorimer, D.R. et al. A bright millisecond radio transient of extragalactic origin. *Science* **318**, 777 (2007).
29. Mandelbaum, R. et al. Galaxy Halo Masses and Satellite Fractions from Galaxy-Galaxy Lensing in the SDSS. *Mon. Not. R. Astron. Soc.* **368**, 715 (2006).
30. Mao, S. and Schneider, P. Evidence for substructure in lens galaxies? *Mon. Not. R. Astron. Soc.* **295**, 587 (1998).
31. Metcalf, R.B. and Madau, P. Compound Gravitational Lensing as a Probe of Dark Matter Substructure within Galaxy Halos. *Astrophys. J.* **563**, 9 (2001).
32. Milgrom, M. A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *Astrophys. J.* **270**, 365 (1983).
33. Nakamura, T.T. and Deguchi, S. Wave Optics in Gravitational Lensing. *Prog. Theor. Phys. Suppl.* **133**, 137 (1999).

34. Navarro, J.F., Frenk, C.S., and White, S.D.M. A Universal Density Profile from Hierarchical Clustering. *Astrophys. J.* **490**, 493 (1997).
35. Paczyński, B. Gravitational Microlensing by the Galactic Halo. *Astrophys. J.* **304**, 1 (1986).
36. Perlmutter, S. et al. Measurements of Omega and Lambda from 42 High-Redshift Supernovae. *Astrophys. J.* **517**, 565 (1999).
37. Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. *Astron. Astrophys.* **641**, A6 (2020).
38. Reasenberg, R.D. et al. Viking relativity experiment: verification of signal retardation by solar gravity. *Astrophys. J. Lett.* **234**, L219 (1979).
39. Refsdal, S. On the possibility of determining Hubble's parameter and the masses of galaxies from the gravitational lens effect. *Mon. Not. R. Astron. Soc.* **128**, 307 (1964).
40. Riess, A.G. et al. Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant. *Astron. J.* **116**, 1009 (1998).
41. Rubin, V.C. and Ford, W.K. Rotation of the Andromeda Nebula from a Spectroscopic Survey of Emission Regions. *Astrophys. J.* **159**, 379 (1970).
42. Sahu, K.C. et al. Relativistic deflection of background starlight measures the mass of a nearby white dwarf star. *Science* **356**, 1046 (2017).
43. Schneider, P. A new formulation of gravitational lens theory, time-delay, and Fermat's principle. *Astron. Astrophys.* **143**, 413 (1985).
44. Schneider, P., Ehlers, J., and Falco, E.E. *Gravitational Lenses*. Springer, Berlin (1992).
45. Sereno, M. Gravitational lensing by spinning gravitational sources. *Phys. Rev. D* **67**, 064007 (2003).
46. Shapiro, I.I. Fourth test of general relativity. *Phys. Rev. Lett.* **13**, 789 (1964).

47. Shapiro, S.S. et al. Measurement of the solar gravitational deflection of radio waves using geodetic very-long-baseline interferometry data, 1979–1999. *Phys. Rev. Lett.* **92**, 121101 (2004).
48. ALMA Partnership. ALMA Long Baseline Observations of the Strongly Lensed Submillimeter Galaxy HATLAS J090311.6+003906 at  $z=3.042$ . *Astrophys. J. Lett.* **808**, L4 (2015). [SDP.81]
49. Thornton, D. et al. A Population of Fast Radio Bursts at Cosmological Distances. *Science* **341**, 53 (2013).
50. Tyson, J.A., Valdes, F., and Wenk, R.A. Detection of systematic gravitational lens galaxy image alignments: Mapping dark matter in galaxy clusters. *Astrophys. J. Lett.* **349**, L1 (1990).
51. van Albada, T.S. et al. Distribution of dark matter in the spiral galaxy NGC 3198. *Astrophys. J.* **295**, 305 (1985).
52. Vegetti, S. and Koopmans, L.V.E. Bayesian strong gravitational-lens modelling on adaptive grids: objective detection of mass substructure in galaxies. *Mon. Not. R. Astron. Soc.* **392**, 945 (2009).
53. Verlinde, E. Emergent Gravity and the Dark Universe. *SciPost Phys.* **2**, 016 (2017).
54. Walsh, D., Carswell, R.F., and Weymann, R.J. 0957+561 A, B: Twin quasistellar objects or gravitational lens? *Nature* **279**, 381 (1979). [First observed quasar gravitational lens]
55. Wambsganss, J. Gravitational Lensing in Astronomy. *Living Rev. Relativity* **1**, 12 (1998).
56. Wong, K.C. et al. H0LiCOW-XIII. A 2.4 per cent measurement of  $H_0$  from lensed quasars:  $5.3\sigma$  tension between early- and late-Universe probes. *Mon. Not. R. Astron. Soc.* **498**, 1420 (2020).
57. Zumalacárregui, M. and García-Bellido, J. Transforming gravity: from derivative couplings to matter to second-order scalar-tensor theories beyond the Horndeski Lagrangian. *Phys. Rev. D* **89**, 064046 (2014).

58. Zurlo, A. et al. The mass of the lens galaxy of HE 0435–1223 estimated through stellar dynamics. *Mon. Not. R. Astron. Soc.* **481**, 1041 (2018).
59. Hanks, M. Phase Theory: Particles as Topological Defects (Paper 3). arXiv:2601.00003 [hep-th] (2026).
60. Hanks, M. Phase Theory: Metric Emergence and the Post-Newtonian Limit (Paper 8). arXiv:2601.00008 [gr-qc] (2026).
61. Hanks, M. Phase Theory: Dark Sectors and Neutral Topological Defects (Paper 12). arXiv:2601.00012 [hep-ph, astro-ph.CO] (2026).
62. Hanks, M. Phase Theory: Phase-Silent Structures and Extended Admissibility Halos (Paper 54). arXiv:2601.00054 [astro-ph.CO] (2026).
63. Abell, G.O. The Distribution of Rich Clusters of Galaxies. *Astrophys. J. Suppl.* **3**, 211 (1958).
64. Meena, A.K. and Bagla, J.S. Gravitational lensing of gravitational waves: Wave nature and prospects for detection. *Mon. Not. R. Astron. Soc.* **492**, 1127 (2020).

---

*End of Paper 45*

---

**Phase Theory Series** • Marlon Hanks • arXiv:2601.XXXXX [gr-qc, hep-th]

Papers 1–100 available at arXiv.org under the Phase Theory series designation (Hanks 2026). Correspondence: [email protected]

This document was prepared using the Phase Theory document preparation standards. © 2026 Marlon Hanks. All rights reserved under the terms of the Dust Open Source License .

Submitted: 16 January 2026 · Paper 45 of the Phase Theory Series

